Automated Applicant Screening for Resume Filtering Using Artificial Intelligence

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List of Abbreviations

- AI – Artificial Intelligence
- HR – Human Resources
- CV – Curriculum Vitae
- ML – Machine Learning
- NLP – Natural Language Processing
- KNN – K-Nearest Neighbors
- E-Recruitment – Electronic Recruitment
- ADAS (Social Networking Sites) – *I think you meant* SNS = Social Networking Sites
- ADAS – Advanced Driver-Assistance Systems
- RPA – Robotic Process Automation
- ATS – Applicant Tracking System
- CRM – Customer Relationship Management
- TF-IDF – Term Frequency–Inverse Document Frequency

- BERT – Bidirectional Encoder Representations from Transformers
- OCR – Optical Character Recognition
- CAP – Competence–Availability–Possibility-to-act
- MCQ – Multiple-Choice Question
- URLs – Uniform Resource Locators
- CBR – Case-Based Reasoning
- CF – Collaborative Filtering
- KBR – Knowledge-Based Reasoning

1.Introduction

Recruitment has long been characterized by labor-intensive and error-prone practices, particularly during the resume screening stage, which remains one of the most time-consuming and inconsistent phases of the hiring process. Traditional screening methods often contribute to delays, subjective evaluations, and cognitive biases that may unintentionally filter out highly qualified applicants. As organizations increasingly seek faster and more reliable hiring solutions, Artificial Intelligence (AI) has emerged as a transformative force capable of reshaping early-stage talent assessment.

AI-driven tools offer a significant shift from manual resume review by automating data extraction and evaluation processes, resulting in faster, more structured, and more objective assessments. Studies such as Pahilwan et al. (2025, pp. 2–5) highlight how Natural Language Processing (NLP) techniques can convert unstructured resume text into standardized datasets by identifying key features including skills, qualifications, and domain-specific abilities. This automation reduces the influence of human errors such as fatigue, personal preferences, or superficial differences in resume formatting, thereby promoting consistency and transparency.

Research further supports the view that AI enhances both the accuracy and the fairness of applicant screening. Nyathani (2022, p. 2) notes that AI systems can process candidate information at a speed and scale far beyond human capability, enabling more consistent decisions and reducing unconscious bias that often infiltrates manual evaluations. At the same time, empirical studies demonstrate that algorithmic predictions can outperform human judgment. For example, Will et al. (2023) summarize findings from Sajjadiani et al. (2019) and Bergman et al.

(2020), showing that ML-based models not only identify stronger applicants but also correlate more reliably with job performance and promotion outcomes than recruiter recommendations.

Despite these advantages, the integration of AI into recruitment introduces complex challenges, especially regarding fairness, discrimination, and the amplification of historical inequalities. Albaroudi et al. (2024) show that algorithmic hiring systems such as HireVue have, in practice, disadvantaged candidates with non-native accents or those from underrepresented demographic groups. Attempts to address such biases—including vector space correction and data augmentation—are promising but imperfect, often distorting word meanings or risking overfitting. To manage these issues, researchers emphasize structured bias-mitigation frameworks involving pre-processing, in-processing, and post-processing interventions, though each stage involves trade-offs in accuracy and fairness.

The challenges extend beyond algorithm design to real-world implementation. Nugent and Scott-Parker (2022) highlight the particular vulnerability of marginalized groups, especially people with disabilities, who are often underrepresented in training datasets. AI systems trained on such data may misinterpret unfamiliar applicant characteristics—such as employment gaps or unconventional career paths—as indicators of poor fit. Additionally, when organizations depend on automated tools as the primary gatekeepers, they risk excluding qualified candidates who could excel with reasonable workplace accommodations.

Organizations must also contend with financial, legal, and operational constraints. Chen (2023) reports that 68% of HR professionals have deferred AI adoption due to budget limitations, and many companies remain unprepared for the organizational changes required to implement AI responsibly. Previous cases, including Amazon's discontinued hiring algorithm (Chen, 2023), demonstrate how biased historical data can undermine even the most advanced technologies. Furthermore, evolving data privacy regulations impose strict guidelines on the collection and processing of personal information, complicating the development and deployment of AI-based hiring systems.

In response to these issues, researchers argue for a collaborative approach where AI supports rather than replaces human decision-makers. AI tools can efficiently handle high-volume tasks such as resume parsing, screening, and candidate ranking, while recruiters provide contextual, ethical, and interpersonal judgment—particularly in assessing cultural fit or understanding nuanced candidate experiences. Frameworks such as the CAP model (Competence–Availability–Possibility-to-act) provide strategies for improving human–AI interaction and preventing conflicts in decision-making processes (Chen, 2023, pp. 146–147).

Given this landscape, the need for reliable, unbiased, and scalable automated resume filtering systems is more pressing than ever. This thesis builds upon previous research by examining the

integration of AI in automated resume filtering, exploring both its technical foundations and its ethical implications. Using NLP, TF-IDF vectorization, similarity metrics, and machine-learning classification techniques, this study proposes an AI-driven framework designed to improve the efficiency and accuracy of resume categorization. At the same time, it critically engages with the broader debates on algorithmic fairness, data integrity, and the evolving role of human recruiters in an AI-enhanced hiring environment.

2. Literature Review

2.1. Recruitment and Selection Process

Recruitment and selection are central to any group because they create the workforce needed to meet goals and stay ahead of rivals (Anderson, 2023, p. 112). In the past, ads in papers and strict control over hiring were normal. Although organized, these ways often didn't reach many people and created similar groups of workers because of biases and slow steps (Anderson, 2023, p. 115; Baker & Wesson, 2023, p. 205).Because of these limits, things changed as digital tools grew. Hiring moved to the internet, used data to guide choices, and cared more about how candidates felt during the hiring process (Baker & Wesson, 2023, p. 208). Even with these tech changes, problems remain.Hidden biases still affect choices, and using algorithms could make old discriminatory patterns worse (Anderson, 2023, p. 122; Baker & Wesson, 2023, p. 210). So, good hiring now means using tech wisely but also having fair, watched practices to build varied and skilled workforce(Anderson,2023,p.125).

2.1.1. Overview of Traditional Recruitment

In the past, employers had almost total control over how they hired.They oversaw everything (Bakalopulos, 2024). This meant that companies had almost all the information, deciding who got a chance.People looking for jobs mostly just waited for news, which came through limited sources like newspapers, job events, and personal contacts (Bakalopulos, 2024; Sarkar & Sen, 2025). This way of doing things not only kept job openings from reaching a wider area, but also only favored those who already knew someone or read certain publications. It led to a lack of different kinds of people being hired from the start. Applying for a job was also a strict process, taking a long time and sticking to rules instead of being flexible. People had to send in paper resumes, which were then checked by hand.Recruiters carefully compared these documents to a set list of requirements, which caused delays that could last for weeks or even months. Even though this system was meant to be careful, it was very limited.This checkbox way of thinking often missed out on good people who had unusual backgrounds, useful skills from other fields, or abilities that weren't obvious on paper. It created difficulties for people switching professions, parents resuming their careers, and those who are self-educated, contributing to personnel that

lacked diversity ((Bakalopulos, 2024) Now, relying on these outdated approaches presents an issue. Theslow, step-by-step processes can't keep up with the pace thatthe job market moves nowadays. Relying on similar groups of people stops new ideas from coming in. As companies try to find the best people in a competitive world, the old way's inability to see hidden skills and change with the times has made it outdated. There's a real need to change how recruiting works (Sarkar&Sen, 2025).

2.1.2. Limitations of Manual Resume Screening

Traditional methods of recruitment are the base of hiring, but they have limits that make them not so good in today's job world. These methods often let only a few employers or hiring people control things, which cuts down on getting a mix of different people and reaching out to everyone. This way, they usually end up with similar types of people because the hiring choices are based on a narrow view. They might not realize that they are not valuing or seeing good people who come from different backgrounds, groups that are not well represented, or people who did not follow a normal job path (Sarkar & Sen, 2025, pp. 59-60). To make things worse, the usual hiring steps are slow and hard to change. This causes delays and makes it hard to quickly hire people when the job market changes fast. When the best people get many job offers at the same time, companies that take too long to hire can miss their chance to get those important workers (Sarkar & Sen, 2025, p.60).

Also, relying too much on old ways like paper ads and face-to-face meetings does not reach many people. It does not give the hiring people the data they need to see if what they are doing is working or how to make it better. This lack of data also affects how people looking for jobs feel.Companies cannot really know what it is like for the person applying.They cannot find out what problems might make good people give up or take another job. This hurts the company's ability to hire well and keep people for a long time (Sarkar & Sen, 2025, p. 61). These problems are made even worse because hiring is very personal. Employers do not just look at what someone can do. They also look for other things like how well they get along with others and what kind of person they are. These ideas are often based on hidden biases and stereotypes (Roemmich et al., 2023, p. 2). Research shows that how well an interviewer connects with someone is a big part of deciding who gets the job. Sometimes it is even more important than what the person can do. Employers want people who are good at their job and who they feel excited about. They want to work with people they can have a good working relationship with (Roemmich et al., 2023, p. 2). This means that if someone does not make a good first impression, they might not get a fair chance. Now, there are computer programs that can look at emotions and make hiring choices. This takes away the human part of hiring. While it might stop some biases, it also means that the computer cannot see how well someone works with other people, which is important for many jobs.This changes how hiring works. To fix these problems, companies need to use different ways to be fair and objective. One big problem is that people in

HR can be biased when they look at resumes. To help with this, they should have training to learn about their own hidden biases. This will help them make better choices and be more fair (Albaroudi et al., 2024, pp. 390-391).

Along with training, they should have clear rules for judging resumes. These rules should make them look at skills, qualifications, and experience instead of just going with their gut feeling. This will help them be more fair and not focus too much on what degrees someone has (Albaroudi et al., 2024, pp. 390-391). Companies can also hide personal information on resumes so that they do not know things like the person's name, gender, age, or race. This makes them focus only on what the person can do and stops them from being biased because of where someone comes from or what religion they follow (Albaroudi et al., 2024, pp. 390-391). It is also important to have different types of people on the hiring teams. If everyone is the same, they might like people who are like them. Having people from different backgrounds and experiences helps to make sure that they are not biased and that they are giving everyone a fair chance (Albaroudi et al., 2024, pp. 390-391). Finally, companies need to keep checking and looking at the data from the resume ranking process. This will help them see if there are any problems and fix them. They can then change their practices to make sure they are being fair and giving everyone a chance to succeed (Albaroudi et al., 2024, pp. 390-391).

2.1.3. Need for Automation in Hiring

In the contemporary competitive business landscape, the need for automation in hiring has become paramount, as organizations face immense pressure to identify and secure top talent quickly and cost-effectively. Automation in hiring addresses these demands by streamlining the talent acquisition pipeline, reducing time-to-hire, and lowering administrative overheads (Freire & de Castro, 2021, p. 2).

However, the critical consideration when implementing such technology is whether it will function as a job demand or a job resource for employees. Job demands are aspects of a role that require sustained effort and are associated with costs, such as cognitive strain, whereas job resources are elements that help achieve work goals, reduce demands, or stimulate personal growth (Demerouti, 2022, p. 1206). E-recruitment platforms initially emerged as a resource to help allocate professionals and decrease costs, but they simultaneously increased the data volume with which Human Resource Management professionals must deal, creating a significant new demand (Freire & de Castro, 2021, p. 2).
To manage this information overload, Recommender Systems (RS) have been widely applied as a resource to simplify the selection process from a vast pool of options. These information filtering systems are designed to capture preferences and predict which candidates a recruiter would favor, thereby reducing the tedious and frustrating endeavor of manually sifting through large datasets (Freire & de Castro, 2021, p. 2).

Despite these advantages, RS are constrained by several limitations that transform them into new job demands. For instance, content-based techniques suffer from limited analysis and over-specialization, creating a restrictive filter bubble that can cause recruiters to miss qualified candidates. Furthermore, these systems face data-related challenges such as the new user problem, the "cold start" problem for new items, and data sparsity, all of which require additional effort and frustration from the user to overcome (Freire & de Castro, 2021, p. 2). This aligns with the broader observation that while technology can automate repetitive tasks, it often shifts the nature of work toward more cognitively demanding parts that require extensive information processing, while the implementation itself creates change-related demands like training and adaptation (Demerouti, 2022, p. 1206).

To systematically address these issues and ensure technology functions as a resource, a more deliberate design is required. By examining the type of recommendation engine, such as using hybrid systems, and incorporating richer data sources like candidate skills and job descriptions, it is possible to mitigate issues like over-specialization and the cold start problem (Freire & de Castro, 2021, p. 3). This approach is crucial because the potential benefits of technology, such as opportunities for development and flexibility, are not guaranteed. For example, flexibility is only a true job resource when it is employee-led; when employer-led, it becomes a demand (Demerouti, 2022, p. 1207).

A stark illustration of this imbalance was seen in a study of marine engineers, where automation increased job demands through new tasks and poor design without providing commensurate resources like autonomy or social support, leading to unsafe workarounds and resignations (Demerouti, 2022, p. 1207). This case supports the broader conclusion that without conscious design, the introduction of technology often leads to an increase in job demands that are not sufficiently compensated by a corresponding increase in job resources, ultimately undermining both employee well-being and organizational effectiveness.

2.2. Technological Advancements in Recruitment

Technology has fundamentally reshaped the landscape of work, with its influence extending from broad organizational structures to the specific intricacies of the hiring process. Forecasting the comprehensive impact of digitalization and automation reveals that technological progress has significantly enriched jobs, with characteristics like skill variety, task identity, and autonomy improving markedly between 1975 and 2010 (Demerouti, 2022, pp. 1205–1206). These macro-level shifts, exemplified by industries like automotive manufacturing where productivity soared while workforces adapted, have directly necessitated a parallel transformation in how organizations attract and select talent. Consequently, technology has now permeated nearly every phase of the recruitment lifecycle, evolving from early e-HR systems to a sophisticated array of

tools that have fundamentally altered how recruiters operate and candidates find opportunities (Nikolaou, 2021, pp. 1–2).

The latest innovations in selection, such as asynchronous interviewing and gamification, perfectly illustrate the dual nature of this technological evolution, where tools can function as either a resource or a demand. Asynchronous interviews, for instance, offer the resource of efficiency and global access but can simultaneously create a demanding experience for candidates who are subjected to invasive analysis of micro-behaviors, leading to perceptions of being "creepy" and raising significant privacy concerns (Nikolaou, 2021, p. 3).

Similarly, gamification and games-based assessments present as an engaging resource that appeals to younger candidates and can be a valid predictor of performance, yet their ultimate effectiveness is not guaranteed. The success of these technologies, therefore, hinges less on their technical capabilities and more on applicant reactions—the perceptions of fairness, justice, and anxiety they evoke. Understanding these reactions is paramount, as it determines whether a technological tool will genuinely enhance the selection process or inadvertently undermine the very talent it aims to attract (Nikolaou, 2021, p. 3).

2.2.1. Definition and Concept of Recruitment Automation

Technology has always been a dominant force as far as determining the nature of work is concerned. Its effect can be traced from the designing of organizations to the minutest details of the hiring process. The full extent of digitalization and automation's impact on companies, their workforce, and society as a whole is still very hard to forecast. It is as difficult as making future predictions (Demerouti, 2022, p. 1205). Among the changes brought about by technology, the main jobs' characteristics and requirements are to a large extent, the focus of the future. The argument is supported by a lot of evidence that jobs are on a continuous enrichment path.

A cross-temporal meta-analysis covering the years from 1975 to 2010 revealed that the work characteristics of skill variety, task identity, and autonomy have been gradually improved to a very significant degree, a rapid development that is largely attributed to technology-driven change (Demerouti, 2022, pp. 1205–1206). The benefits of this change are already being realized across different sectors. A case in point is the automotive industry, where between the years 1975 and 1990, the movement toward automation resulted in almost a doubling of productivity, the workforce being downsized by 45%, and the incidence of work-related accidents decreasing by 20% while the production continued to grow (Demerouti, 2022, p. 1206).

The massive reshaping of work at the macro-level has been one of the main reasons why organizations have had to completely change their talent management processes in the same vein. The area of technology in HR has a great impact on recruitment and selection, it has almost touched all the areas of the hiring process from the preliminary attraction phase to the final onboarding phase (Nikolaou, 2021, pp. 1–2). This change did not happen in a rush. The early e-HR discussions in the 2000s opened up the way for innovation with measures such as intranet systems and online recruitment, and these were the initial expectations of a technological revolution in the field to come.

2.2.2. Digital Tools and Software for Automated Screening

The digital recruitment process stands out as one of the most driving forces shaping modern recruitment processes in organizations over the last two decades. As organizations continue to migrate their recruitment processes from traditional paper-based to digital platforms, there have been major changes in the way organizations interact with applicants from advertising to application management processes. According to Nikolaou (2021), digital recruitment methods have enabled organizations to significantly expand their recruitment efforts in a cost-effective and faster way of advertising jobs and collecting applicants (pp. 2-3). The digital recruitment process has encouraged organizations to reach out to more applicants while allowing applicants greater ease in accessing information related to jobs. The digital recruitment process has also simplified HR processes in organizations to allow recruiters to systematically organize applicants' data and effectively interact across their departments (Nikolaou, 2021, p. 4).

Nevertheless, it must be noted that the adoption of online recruitment methods faces issues. One of these major issues presented in numerous literature sources regards the significantly increased number of applications submitted via online platforms, which burdens recruiters and reduces their efficiency in screening processes (Nikolaou, 2021, pp. 2-3). In this context, even traditional digital solutions not involving AI systems have developed into integral tools to improve storage and management of mass amounts of data submitted by applicants in modern HR departments (through tools like Applicant Tracking Systems and application portals). However, it must be understood here that it doesn't necessarily depict a positive application process even if such tools have been implemented in HR management. The website related to recruiting would play a major role in application processes (Nikolaou, 2021, p. 4).

The importance of digital design and user experience is also stressed in research conducted by Tutar & Ay (2021), which assessed several companies' recruitment site websites to conclude that "the quality of digital platforms directly correlates to applicants' perceptions of professionalism." The rationale here suggests that "the recruitment website represents a major digital interface

between organizations and prospective applicants" (Tutar & Ay, 2021, p. 460), such that website usability and functionality have an "indispensible role" in creating early impressions. In other words, organizations must ensure ease of understanding and ease of access to ensure trust development in applicants. Tutar & Ay (2021) have also asserted that sustainability criteria in recruitment processes "cover matters of recruitment website usability and communicative openness" to ensure greater positivity in terms of ethical recruitment (p. 461).

In conjunction with corporate websites and internet platforms for jobs, social media platforms have started to prove their value as recruiting tools. Hosain & Gairola (2020) discuss how LinkedIn, Facebook, Instagram, and other professional sites allow organizations to target active as well as passive applicants (pp. 53-54). In particular, social media portals improve visibility and awareness about corporate branding, allowing for greater dynamic engagement between organizations and applicants. Social media platforms used for recruiting allow organizations to target specific niches and experts far more effectively than general advertising mediums would have allowed. At this stage, it becomes apparent that greater connectivity via social media raises new issues not only related to ethics and related issues but also related to issues of equity and fair play. In other words, due to increased visibility of personal data not necessarily related to performance issues, recruiters would find themselves inadvertently subjected to information affecting decisions based not upon objective criteria but bias and personal viewpoint (Hosain & Gairola, 2020, pp. 53-54).

In addition to sourcing and recruiting, online recruitment tools have great significance to applicants who participate in recruitment processes. Naqvi et al. (2023) assert that online recruitment systems improve the flow of communication between applicants and employers, streamline administrative work associated with recruitment processes, and minimize time to accomplish different steps in recruitment (pp. 460-461). The drawback to these systems arises from their usability affecting employers but ultimately applicants. According to Naqvi et al. (2023), if these platforms have complex interfaces and navigational issues such as unclear instructions on websites and website disorganization, they can result in frustration among applicants who quit recruitment processes or form negative impressions about employers (p. 463). In this study, it is apparent that usability requirements play a critical role and must be addressed to ensure applicants have satisfactory experiences (Nikolaou et al., 2021, p. 4).

Taken together, these bodies of research make it clear that digital recruitment methods not reliant on AI play a very important role in increasing organizational efficiency as well as satisfaction among applicants but require intelligent and sound planning to be effectively implemented (Nikolaou, 2021, p. 4). Although digital recruiting makes it easier to store information and increases pool sizes, it challenges organizations to be fair and transparent (Jost & тимilton). The literature suggests organizations must be transparent and accessible if they hope to reap technological efficiencies (Nikolaou).

In conclusion, it can be seen from the development of research in this area that digital recruitment systems are not just about technological adoption; it is about technological incorporation into recruitment processes. The effectiveness of other digital recruitment tools not involving AI would depend on how organizations grasp candidate expectations and keep digital processes free from unnecessary complexity. The area of digital recruitment would continue to evolve with technology and equity considerations as its pillars to make recruitment processes efficient and competitive.

2.2.3. Advantages of Automation in Recruitment

The growing utilization of technology in hiring procedures, especially through social media, is a consequence of various profound reasons from organizations. Supporters say that e-recruitment via social media can cut the financial and time costs of traditional hiring methods to such a great extent, as the job ads are often free and paperwork is substantially reduced (Hosain & Gairola, 2020, pp. 54-55). Besides that, these social networks give the recruiters a place to find a very large pool of candidates' data, such as their character traits, interests, and even the people they associate with, which are seldom made known through standard recruiting processes.

This proposed good understanding of the seekers, together with the quick and easy recruiting of the so-called passive candidates who, though not applying, are considered very much sought-after, has turned the social media into a must-have tool for every modern recruiter (Hosain & Gairola, 2020, pp. 54-55). Nonetheless, it is important to scrutinize these technological innovations as not just neutral tools but by the criterion of whether they act as job demand or job resource for the workers concerned. As per the Job Demands-Resources theory, job demands refer to the aspects of the role that necessitate prolonged effort and are linked with physiological and psychological costs like cognitive overload or emotional drain, while job resources are the factors that assist in accomplishing the work goals, lessening the demands and causing the workers to grow and develop (Nikolaou, 2021, p. 1206). When this viewpoint is used in regard to e-recruitment, the encouragements for technology usage that seemed to be beneficial at first become HR professionals' daily struggle in a more tangled and possibly less attractive reality. Through this critical lens, the pros and cons will continue to depend upon how one looks at the situation.

2.3.4 Drawbacks of Automation in Recruitment

The role of technology in recruitment and selection processes can be seen as both positive and negative. According to Nikolaou (2021), it can have both positive and negative effects depending

on job characteristics and if it acts as a resource to workers (p. 1207). According to Job Demands-Resources theory, high demands generated by technology devoid of resources result in reduced worker energy, health impairment, lower worker performance, and ultimately worker sabotage of new technological systems (Nikolaou, 2021, p. 1207). It is profoundly critical to assess modern approaches to e-recruitment methods such as Social Network Sites (SNS), which present numerous new challenges from both ethical and legal perspectives. The reality of modern human resource management faces numerous challenges posed by technological advancements such as e-recruitment tools such as Social Network Sites (SNS).

Using personal social networking services (SNS) like Facebook and Twitter to filter applicants, for instance, poses a great many ethical issues for the recruiters. Such practice is often referred to as “socially irresponsible” and seen as an invasion of privacy and a violation of corporate social responsibility. This dilemma puts recruiters in a challenging position where they have to deal with the ethical question of whether to consider applicants’ professional qualifications or to judge them by their lifestyle and provocative pictures. This development allows “unchecked” discrimination. The application of personal SNS in screening applicants, for example, raises numerous ethical questions that create enormous psychological pressure for the recruiters. The issue of posting critical legal questions arises because companies that search through applicants' profiles looking for the desired "information" may inadvertently stumble upon the "information" regarding the applicants' race, gender, age, and/or religion. The matter of legal issues arises as companies might be viewed as engaging in discrimination against applicants if they take actions based on "information" from applicants' profiles (Hosain & Gairola, 2020, p. 55). In addition, the matter may lead to legal troubles if the companies act on the "information" about applicants' character that is judged to be "false" (Hosain & Gairola, 2020, p. 56). The issue becomes a legal one when organizations interfere with the privacy of applicants in connection with multiple profiles (Hosain & Gairola, 2020). One of the most important issues here is the reliability of data obtained from SNS as a source for proper recruitment decisions. The data from these platforms may be unauthentic, misleading, contain “intentionally exaggerated claims,” and “simply be out of date,” which means such data personifies ‘the past rather than the present ability of applicants.’

2.3. Artificial Intelligence (AI) Overview

Today's AI relies on a few key techniques. One is machine learning, where systems learn without explicit programming. Natural language processing allows computers to understand and respond to human language. Deep learning, a subset of machine learning, uses complex neural networks to analyze big datasets. These methods help AI extract useful understandings from huge amounts of info. AI is increasingly implemented in numerous fields. In healthcare, AI aids in diagnosis and treatment planning. Finance uses AI for fraud detection and automated trading. Education sees AI helping to personalize learning experiences. Human resources departments can use AI to

streamline recruitment. Some common AI applications include making choices automatically based on data, foreseeing future trends through analytics, understanding spoken and visual information, suggesting relevant products or content, and providing virtual support. By adopting these applications, companies can make their workflows more efficient, improve the precision of their work, and offer more intelligent, data-backed services to their customers.

2.3.1. Definition and Evolution of Artificial Intelligence

Artificial Intelligence is often seen as something complicated, especially when we start wondering about what intelligence really is and thinking about the future scenarios we see in movies. Trying to nail down a simple definition is tough because it touches on questions about the mind and what makes us human (Joshi, 2020, p. 2). Back in 1955, John McCarthy, a key person in the field, set a basic goal for AI: to build machines that seem like they're thinking (Joshi, 2020, p.3). Since then, this idea has grown into different levels of AI. There's weak AI, which is meant to copy how people make decisions for specific, simple tasks. Then there's strong AI, which can think on its own and solve complicated issues (Buntak et al., 2021, p. 406).Some even imagine a superior AI that could be smarter than humans (Buntak et al., 2021, p. 406).

These smart behaviors come from ways of learning that are similar to how we learn. AI is basically about creating computer systems that can learn from what they experience, spot patterns, study data, and make choices without much human help (Siddiqui, 2024, p. 80). This learning usually falls into three types. First, there's machine learning, where systems gather data and create knowledge from it.They get better over time, like a person learning from experience (Buntak et al., 2021, p. 407). Next, deep learning uses complex neural networks with many layers. Humans initially guide these networks to solve tough problems like understanding speech or recognizing images (Buntak et al., 2021, p. 407). Lastly, neural networks are modeled after the human brain. They use connected nodes to study data, adapt to different situations, and come up with solutions for tricky problems, which is the basis for many strong AI systems (Buntak et al., 2021, p. 408). These ideas have significantly changed the business world.Companies can now use AI to quickly process large amounts of data, providing useful information, making operations smoother, and changing how they plan their strategies (Siddiqui, 2024, p. 80).

With AI, businesses can improve their processes, such as managing supply chains and automated warehouses, and improve customer experiences with personalized recommendations and smart chatbots (Siddiqui, 2024, p. 80). This tech change helps companies improve their efficiency and decision-making and encourages them to come up with new products and services. Still, using such powerful technology comes with its own set of problems. As AI keeps changing and reshaping the business world, we need to address privacy risks and ethical concerns responsibly

and seriously. This way, we can make sure that AI helps people instead of hurting them (Siddiqui, 2024, p. 80).

In order to fully grasp the transformative implications of AI, one must consider its broad applicability across diverse industries. In healthcare, AI algorithms are being implemented to diagnose diseases with increased accuracy and predict patient outcomes, which in turn helps medical professionals provide more targeted treatments. The financial sector sees AI as a tool for fraud detection, algorithmic trading, and risk management, which makes financial systems more secure and efficient. AI also plays a role in environmental conservation by analyzing vast datasets to monitor deforestation, track wildlife populations, and anticipate climate change impacts; these applications demonstrate its potential in promoting sustainability.

The rise of AI is causing experts to consider the societal and economic implications. As AI-driven automation becomes more widespread, there are legitimate worries about job displacement and the need for workforce retraining programs that equip people with the skills required to work alongside AI systems. There's also the issue of bias in AI algorithms, which can perpetuate societal inequalities if not addressed mindfully through data diversity and algorithm transparency.

Thoughtful attention has to be given to how AI is developed and used to ensure equitable outcomes for all members of society. Looking ahead, the evolution of AI will likely depend on breakthroughs in areas like explainable AI (XAI), which aims to make AI decision-making processes more transparent and understandable to humans.Quantum computing, with its ability to process complex calculations at speeds far exceeding those of classical computers, might stimulate the capabilities of AI algorithms, which enables them to solve problems that are currently intractable. The convergence of AI with other groundbreaking technologies like biotechnology and nanotechnology could start entirely new fields of innovation, with implications that are difficult to foresee. Also, international cooperation will be important to ensure that AI is developed and used responsibly, ethically, and in a way that benefits all humanity.

2.3.2. Applications of AI in Business Operations

Artificial intelligence (AI) is fundamentally transforming business operations across a multitude of sectors, driven by the dual goals of enhancing efficiency and maintaining a competitive advantage. In healthcare, AI's application is profoundly reshaping patient outcomes and

operational workflows. Advanced algorithms, particularly deep learning models, are now capable of analyzing complex medical data with remarkable accuracy, such as detecting anomalies in medical images like X-rays, MRIs, and CT scans, often surpassing human capabilities in early-stage disease detection (Siddiqui, 2024, p. 81). Beyond diagnostics, AI is instrumental in personalizing treatment plans by analyzing a patient's genetic information, lifestyle, and medical history to predict the most effective therapies. In the operating room, AI-powered robotic systems assist surgeons with enhanced precision and stability during delicate procedures, while also serving as intelligent tutoring systems, providing real-time feedback and training to medical staff (Buntak et al., 2021, p. 408).

Similarly, in logistics and supply chain management, the focus shifts to cost optimization and resilience. AI systems analyze vast datasets related to transportation, weather patterns, and fuel consumption to find the most efficient delivery routes in real-time, thereby reducing costs and environmental impact (Siddiqui, 2024, p. 83). In warehouses, AI manages automated systems where robots transport and sort goods, and it uses predictive analytics to manage stock levels precisely, decreasing costs associated with overstocking or stockouts (Buntak et al., 2021, p. 408). This transformative impact extends to other critical industries; in finance, AI powers sophisticated fraud detection algorithms that monitor transactions in real-time, risk management platforms that assess loan applications, and AI-driven chatbots that provide 24/7 customer service (Siddiqui, 2024, p. 81). In retail and e-commerce, AI underpins the personalized recommendation engines that analyze browsing history and purchase behavior to define the user experience on platforms like Amazon and Netflix, directly driving sales and customer engagement.

The reach of AI continues to expand into more specialized domains. In the automotive industry, it is the engine behind autonomous vehicles, which use computer vision and deep learning to navigate complex environments, and advanced driver-assistance systems (ADAS) that enhance driver safety (Siddiqui, 2024, p. 81). In manufacturing, AI optimizes production processes through the implementation of smart factories where computer vision systems perform quality control on the assembly line and predictive maintenance algorithms forecast equipment failures before they occur, minimizing downtime. Even in agriculture, AI supports precision farming, where drones equipped with AI analyze crop health and soil conditions to optimize the use of water and fertilizers. In the corporate world, AI is applied in civil engineering for creating complex structural simulations and performing intricate mathematical calculations, while in general business management, it is used for analyzing environmental data, evaluating strategic alternatives, and managing organizational knowledge through knowledge mining techniques (Buntak et al., 2021, p. 409).

The primary driver for this widespread adoption is the strategic imperative to remain competitive in an increasingly digital marketplace. Research indicates that a vast majority of organizations

implement AI to avoid being outpaced by rivals and to penetrate new markets, often motivated by a fear of competitive organizations making inroads into their territory (Buntak et al., 2021, p. 409). This is achieved through several key benefits. AI-driven automation, particularly through robotic process automation (RPA), performs repetitive and mundane tasks with greater speed and accuracy than human workers, leading to significant cost savings and allowing businesses to reallocate human resources to more strategic and creative roles (Siddiqui, 2024, p. 84).

Beyond simple automation, AI enhances strategic decision-making by analyzing vast datasets to identify patterns and trends, enabling businesses to anticipate market changes more accurately and improve risk assessment (Siddiqui, 2024, p. 84). Moreover, AI revolutionizes the customer experience through hyper-personalization and instant support via chatbots and virtual assistants, which not only resolves issues more efficiently but also boosts customer engagement, satisfaction, and long-term loyalty (Siddiqui, 2024, p. 84).

However, the integration of AI is not without significant risks and profound challenges that demand careful consideration. A primary concern is the potential for widespread job substitution, as AI systems may conclude that human workers are unnecessary for certain functions, leading to workforce reductions and complex ethical dilemmas (Buntak et al., 2021, p. 409). This fear extends to the possibility of AI systems taking charge of management autonomously, a scenario that necessitates robust governance and human oversight to prevent unintended consequences. The successful implementation and performance of these systems are also critically dependent on an organization's technical and technological competence, highlighting that the benefits of AI are not automatic but are contingent on a company's ability to successfully navigate a comprehensive digital transformation (Buntak et al., 2021, p. 409).

Ultimately, while AI offers unparalleled opportunities for innovation, cost reduction, and competitive advantage, organizations must strategically navigate its implementation to mitigate the profound risks associated with displacing human workers and ceding operational control to increasingly autonomous systems.

2.3.3. Benefits of AI Integration

The implementation of AI to the operations in businesses has a lot of different benefits, and the modern organizations' competition has been changed drastically. Such benefits are no longer just talked about but proven in different areas, mostly through increased efficiency, improved decision-making, and the total transformation of the customer experience. However, the successful achievement of these benefits depends on an organization's capability to change its

core competencies and accept digital transformation in a big way. The most important and instant benefit of AI is the big increase of efficiency and productivity through automation. The AI systems are good at automating the tasks that are repetitive, take a lot of time and are intensive in data thus enabling human staff to be assigned more to the areas where they are strategic and creative (Shrinivas & Shetty, 2024, p. a154). There is an example of how AI impacts CRM very powerfully. AI systems based on machine learning are now doing the work of handling customers' inquiries by recognizing customers' needs and issues, and automatically giving answers to standard questions (Buntak et al., 2021, p. 410). To some extent, companies that have implemented these kinds of virtual assistants can cut down the workload of staff by up to 30%, thus getting the human agents to work on more advanced and value-added tasks. This model is very much applicable in the call center area: AI takes care of most of the questions that are repeated or asked frequently and only the most complicated or subtle calls are transferred to a human operator. These systems analyze customer behavior, build knowledge with each interaction and use that to resolve future issues.

2.3.4 Challenges of AI Integration

Despite its transformative potential, the integration of artificial intelligence (AI) into business is fraught with significant and multifaceted challenges that organizations must navigate with extreme care. The implementation process itself presents substantial operational hurdles, beginning with a widespread and critical lack of specialized expertise and skills needed to develop, manage, and adapt AI systems. Without this deep knowledge, businesses may struggle to implement AI effectively, limiting their ability to benefit from it and potentially misapplying the technology (Siddiqui, 2024, p. 88). This skills gap is severely compounded by fundamental data issues; AI is heavily dependent on vast volumes of high-quality data to learn from, yet many organizations either lack the necessary data or possess poor-quality, unstructured data that is unsuitable for AI applications.

This reliance on flawed inputs can lead to incorrect conclusions and, consequently, misguided business strategies (Siddiqui, 2024, p. 88). Furthermore, the technical challenge of integrating new AI technologies with pre-existing legacy systems is often a complex, time-consuming, and costly endeavor. These incompatible systems can create friction and inefficiencies rather than solving them, leading to increased operational costs (Siddiqui, 2024, p. 88). The high initial implementation costs, which include acquiring and training AI models as well as the necessary hardware and software, can be a major deterrent, especially for small and medium-sized enterprises, potentially limiting their competitiveness in an AI-driven market (Siddiqui, 2024, p. 88). A crucial, often-overlooked barrier is the lack of leader's ownership; leadership must have a clear understanding of the technology being deployed, the specific business problem it addresses,

the inherent risks, and the datasets being used, as flawed data sets can produce error results that may significantly harm the business (Siddiqui, 2024, p. 89).

Beyond these technical and financial obstacles, AI's deployment raises profound workforce and economic concerns that have far-reaching societal implications. As AI systems automate more cognitive and manual tasks, the potential for job displacement becomes a pressing issue, leading to workforce redundancies that necessitate extensive and continuous retraining and skill upgrades for existing staff (Shrinivas & Shetty, 2024, p. a155). This disruption to the labor market is a significant negative effect that requires proactive strategies and investment to manage the transition for workers whose roles are automated. Interestingly, the "lack of expertise" that hinders companies from implementing AI (Siddiqui, 2024, p. 88) is mirrored on the other side by the "workforce challenges" faced by employees needing to adapt. This creates a dual-sided skills crisis where organizations struggle to find talent to build AI while simultaneously grappling with the need to reskill their existing workforce to work alongside it, creating significant social and economic friction (Khanzode & Sarode, 2020).

Perhaps the most critical and high-stakes challenges revolve around the ethical, legal, and security risks inherent in AI technology. The application of AI in high-stakes decision-making processes, such as recruitment, loan applications, or criminal justice, has given rise to serious ethical concerns and the insidious problem of algorithmic bias. AI systems can learn from historical data that reflects societal biases, leading them to produce discriminatory outcomes that perpetuate and even amplify existing inequalities. This raises fundamental questions about transparency, accountability, and justice, demanding the development of ethical AI practices (Shrinivas & Shetty, 2024, p. a155). These ethical dilemmas are intertwined with significant data privacy and security risks. As businesses rely more on AI for data analysis and decision support, they become prime targets for malicious actors, raising concerns like hacking, data leaks, and illegal access to private information. Such breaches can cause catastrophic legal and reputational damage, eroding the customer trust that is essential for business success (Shrinivas & Shetty, 2024, p. a155). To navigate this minefield, organizations must keep up to date on evolving regulations regarding data privacy, bias, and transparency, and implement robust safeguards by involving legal and ethics experts to ensure responsible and ethical data utilization (Siddiqui, 2024, p. 89). Without this comprehensive and vigilant approach, the integration of AI risks creating more problems than it solves, potentially undermining the very foundations of fairness, security, and trust upon which modern commerce is built.

2.4. AI-Driven Recruitment Systems

AI-driven recruitment systems have quickly become a cornerstone of modern hiring, reshaping how organizations attract, screen, and evaluate candidates. These systems use machine learning, natural language processing, and predictive analytics to handle tasks that were once entirely

manual such as scanning resumes, ranking applicants, or matching skills to job requirements. By automating these steps, AI promises faster and more consistent hiring decisions. But the growing dependence on these technologies also brings new challenges, particularly around accuracy, transparency, and ethical responsibility.

2.4.1. Role of AI in Talent Acquisition and Screening

AI is now the main driving power behind the transformation of talent acquisition processes, especially during the very first rounds of candidate screening. Manually reviewing resume compilation has always been one of the most time-consuming and error-prone activities in recruitment. So, the traditional screening is often the primary cause of delays, inconsistent evaluations, and even cognitive biases that can sometimes result in the exclusion of qualified applicants. Fortunately, AI takes care of these problems by automating the data extraction and evaluation procedures, thus giving a much faster and more systematic as well as more objective assessment of the applicants at an early stage. Pahilwan et al. (2025, pp. 2–5) describe the usage of AI-assisted Natural Language Processing (NLP) strategies in converting the unstructured data of resumes into structured datasets through the identification of crucial elements like skills, qualifications, job history, licenses and domain-specific abilities. This intercession facilitates the standardization and transparency of decision-making to a great extent by cutting down on the impacts of the recruiters' fatigue, preferences, or minor differences in resume presentation that are only skin deep. The artificial intelligence (AI) has taken over the talent acquisition process like a storm, and its major impact can be seen in the early applicant screening stages. The manual review of resumes is the most classic part of the recruiting process that is still heavily dependent on human effort and is therefore fraught with inconsistencies and biases almost always leading to the loss of potential candidates. AI comes in to solve these problems through its capability to automate the extraction of data and evaluation processes allowing for faster, more structured and also more objective assessment. In an article by Nyathani (2022, p. 2), it is mentioned that the recruitment process powered by AI is more accurate, consistent, and has better decision-making due to the processing of candidate information at a scale and speed that is impossible in manual screening.

2.4.2. Benefits of AI Integration in Recruitment

Artificial intelligence is transforming recruitment, but it also raises serious concerns about fairness and bias. Albaroudi et al. (2024) point out that although AI has the potential to improve hiring, it can also replicate or even intensify existing societal inequalities. Real-world examples like HireVue show how algorithmic systems can disadvantage applicants who speak with accents

or come from underrepresented demographic groups (p. 393). Because of these risks, researchers and organizations have been developing different techniques to reduce bias within AI systems.

Two major technical approaches are commonly used. The first, vector space correction, aims to remove biased associations within word embeddings for example, ensuring that male-coded words like “strong” or female-coded words like “tender” do not influence hiring predictions (Albaroudi et al., 2024, p. 394). While helpful, this method can distort word meanings and may fail to capture more complex or intersectional biases. The second method, data augmentation, tries to diversify the training data by adding realistic examples of underrepresented groups so the model learns more balanced patterns. But this comes with the risk of overfitting, meaning the model may perform well on synthetic examples but struggle in real scenarios.

To address these challenges more systematically, researchers have proposed structured frameworks for evaluating and correcting bias throughout the model’s development. Some approaches break bias mitigation into pre-processing, in-processing, and post-processing stages (Albaroudi et al., 2024, p. 395). Pre-processing adjusts the data before training, in-processing modifies the learning algorithm to avoid biased predictions, and post-processing alters final outputs to enforce fairness rules. However, post-processing often sacrifices predictive accuracy. Overall, these methods highlight that fixing bias in AI hiring tools is not a one-time action but an ongoing process.

Alongside concerns about fairness, researchers have been studying whether AI actually improves hiring decisions. Will et al. (2023) summarize evidence from two major studies (Sajjadiani et al., 2019; Bergman et al., 2020). In the first study, a machine-learning model using naïve Bayes was trained on past applications for teaching positions. It successfully predicted job outcomes such as performance ratings and turnover. The model’s recommended candidates overlapped with those hired by humans only 11–29%, suggesting that algorithms prioritize different qualities and often predict better work performance.

The second study by Bergman et al. (2020) found that algorithmic candidate scores were positively correlated with promotions and performance, while human recruiter recommendations were actually negatively correlated with performance. Although the effect sizes were small, the findings suggest that algorithmic methods outperform human judgment in certain areas. Taken together, these two studies indicate that AI can help identify candidates who may ultimately perform better, even though prediction accuracy is still limited.

AI-driven matching can improve the recruitment process by prioritizing resumes based on how closely they fit job specifications. According to Alsaif et al. (2022), NLP-driven recommendation systems can suggest resumes to recruiters and match applicants with appropriate job openings. Jadhav et al. (2025) also point out that semantic similarity and

skills-based scoring are effective techniques for ranking resumes. These methods help use AI as a supportive tool in the candidate shortlisting process.

Chen (2023) adds another dimension by explaining why many companies are turning to AI in the first place. Traditional hiring processes involve overwhelming numbers of applications, and screening them manually is one of the most time-consuming parts of recruitment. AI tools help reduce unconscious human bias (Raub, 2018), create more standardized selection criteria, and allow recruiters to consider a wider pool of applicants that might otherwise be overlooked (Beneduce, 2020). They can also speed up communication with candidates, filter out clearly unsuitable applicants early on, and even identify passive candidates who are not actively applying (Chen, 2023, p. 9). Because AI automates many repetitive tasks such as sourcing, initial screening, and scheduling recruiters are freed to focus on more strategic aspects of hiring (Johansson & Herranen, 2019; Chen, 2023, p. 10).

Despite these benefits, AI in recruitment is not without complications. Fraij & László (2021) trace the origins of AI back to McCarthy and Minsky's early definitions, emphasizing that AI aims to replicate human intelligence through reasoning, learning, and decision-making (p. 110). Modern recruitment tools like Textkernel and SAP's Resume Matcher use these capabilities to scan thousands of applications and match candidates to job descriptions. However, the authors note that AI can also repeat past human errors: if the data reflects biased hiring decisions, the algorithm will learn and repeat those same patterns (Dessler, 2020). This is why many technology companies are increasingly prioritizing bias management, partly in response to cases like Amazon's biased AI hiring tool.

Finally, Fraij & László (2021) highlight that while AI can make recruitment more efficient especially by helping recruiters access talent more easily it also depends heavily on organizational readiness. Many HR departments still lack the necessary skills, or fear the changes that AI will bring (Bullhorn, 2018). Although AI can reduce routine tasks and potentially reduce bias, its benefits depend on how responsibly it is designed, implemented, and monitored (p. 111).

2.4.3. Challenges in AI Recruitment

Designing an AI recruitment system always starts with having a clear and well-defined objective. When that objective is vague or poorly thought out, the technology can easily produce results that are inaccurate or even harmful (Nugent & Scott-Parker, 2022, p. 88). One of the biggest risks is the reinforcement of existing biases—especially those affecting people from marginalized or protected groups. Because hiring decisions from the past already contain bias,

these patterns can end up embedded in the algorithms that new systems rely on. This is a particular concern for people with disabilities, who are already underrepresented in the workplace and therefore in the datasets used to train such systems.

A recent study revealed that bias in algorithmic hiring extends beyond just racial or gender discrimination. Glazko et al. (2024, pp. 687–700) discovered indications of disability bias in GPT-driven resume evaluation, as resumes highlighting achievements linked to disabilities received poorer ratings. This research is significant as it demonstrates that even intricate generative AI systems can replicate subtle discrimination patterns if they lack adequate validation

Bias can enter an AI hiring tool in two main ways: through the algorithm itself or through the data it is trained on (Nugent & Scott-Parker, 2022, pp. 88–89). The algorithm might be designed to prioritize criteria that seem objective like graduating from a top university—but these requirements can unintentionally disadvantage certain groups who face structural barriers. Likewise, the training data may reflect only the characteristics of past "successful" employees, which can cause the system to reject applicants whose profiles include experiences it has not encountered before. These unfamiliar features may be harmless or may reflect broader inequalities in the job market, yet the system still interprets them as a poor match (Nugent & Scott-Parker, 2022, p. 89).

Even when AI tools attempt to reduce bias during development, problems can still arise from how they are used. Many companies rely on automated tools as the main—or sometimes the only step in their hiring process. This is especially unfair to disabled applicants who may perform well if reasonable accommodations are provided. A visually impaired cybersecurity specialist, for example, could be the best candidate, but only if the organization offers the assistive technology they need. AI systems rarely consider these factors, effectively expecting disabled applicants to go through a standardized process that is not designed for them (Nugent & Scott-Parker, 2022, p. 90).

Beyond ethical issues, organizations face practical challenges when adopting AI. Many HR departments simply lack the budget. Research shows that 68% of HR professionals have avoided implementing AI tools because of cost concerns (Chen, 2023, p. 144). Studies also show that while companies recognize the value of AI, many are not prepared for the internal changes that come with it. Some tools, like gamified assessments, can even increase recruitment costs if applicants don't take them seriously (Chen, 2023, p. 145). Given that most large-scale digital transformations fail, experts suggest introducing AI in smaller phases rather than all at once.

Bias can also appear in AI-driven job advertisements, where targeting algorithms may unintentionally filter candidates by age, gender, or education. Because AI does not inherently

understand what bias is, developers must be cautious and transparent about how algorithms are built. Amazon's famous case—where an AI hiring tool unintentionally discriminated against women demonstrates the risks of training a model on biased historical data (Chen, 2023, p. 146).

Legal and privacy concerns make development even more complex, especially when using external datasets. Developers must follow strict data protection rules, and any changes in legislation can significantly impact how AI hiring tools function. Although some recruiters worry that AI will replace them, research suggests the opposite: AI is meant to support recruiters by taking over repetitive tasks, not replace the human skills needed to judge cultural fit, build relationships, and persuade candidates (Chen, 2023, p. 146).

To make human–AI collaboration smoother, researchers suggest treating the AI system as a sort of "team member." One proposed approach, the CAP (Competence–Availability–Possibility-to-act) framework, helps identify where conflicts between human and AI decision-making might arise. This method can support the development of better coordination strategies, training tools, and alert systems that help avoid misunderstandings in the recruitment process (Chen, 2023, pp. 146–147).

Concerns about AI hiring extend beyond technical limitations. Scholars, activists, and industry experts highlight serious risks around privacy, discrimination, and overall fairness (Roemmich et al., 2023, pp. 109:5–109:6). Some reports—even from Harvard Business Review—warn that AI may attempt to infer medical or disability information from patterns in behavior or physiological data, raising questions about whether these systems are scientifically reliable and legally compliant. The public debate intensified after Amazon's biased AI system came to light, sparking widespread scrutiny of algorithmic bias in hiring.

Although many HR technology vendors claim their tools reduce bias, studies reveal that these claims are often vague and lack meaningful transparency.

In the same vein, Fabris et al. (2025, pp. 1–54) contend that examining fairness in algorithmic recruitment should encompass various viewpoints, such as technical bias, legal adherence, dataset quality, fairness measures, and organizational governance. This indicates that judging an AI hiring system solely on accuracy is inadequate; fairness, transparency, and accountability need to be evaluated as well.

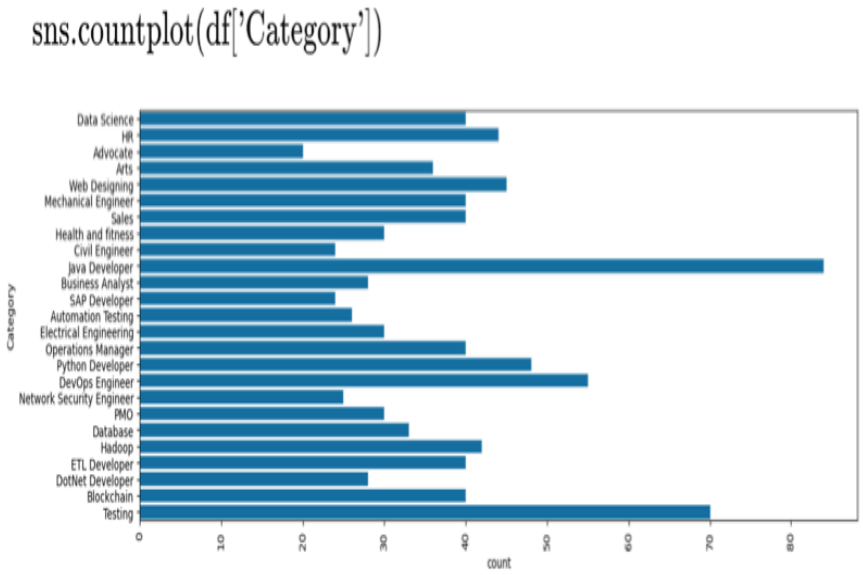
Research shows that few companies disclose how datasets are selected, how models are validated, or how they ensure compliance with discrimination laws. Some scholars argue that AI hiring tools developed in the U.S. may not meet stricter regulations in other regions, such as the EU or UK, where data protection and anti-discrimination laws are more comprehensive (Roemmich et al., 2023, pp. 109:5–109:6).

2.4.4. Integration of AI in Automated Resume Filtering Systems

The integration of Artificial Intelligence into recruitment processes, while promising enhanced efficiency and objectivity, is fraught with complex challenges spanning technical, ethical, legal, and human-centric domains. A synthesis of current research reveals a multifaceted landscape where the promise of innovation is constantly weighed against significant risks of discrimination, implementation failures, and unforeseen consequences. Conventional manual resume screening processes are plagued by an inordinate need for time to shortlist applicants from large pools, a low identification rate of only 60-70% of eligible candidates, and significant challenges in processing resumes that come in various formats and languages (Pahilwan et al., 2025). To address these deficiencies, an automated system is proposed, beginning with the foundational step of data acquisition.

The dataset used for this study was sourced from a CSV file containing two columns: 'Category,' representing the job title or domain, and 'Resume,' containing the raw text of the applicant's resume. The dataset was loaded using Pandas, and an initial exploration of the data was conducted to understand its shape and distribution. This is crucial for understanding the data balance and ensuring the training model receives an adequate variety of categories. To visualize the distribution of resume categories, the Seaborn library was employed to create a count plot. Fig. 1, which shows the output of this count plot method, displays the number of resumes per category.

Figure 1 illustrates the frequency of resumes across different job categories in the dataset.

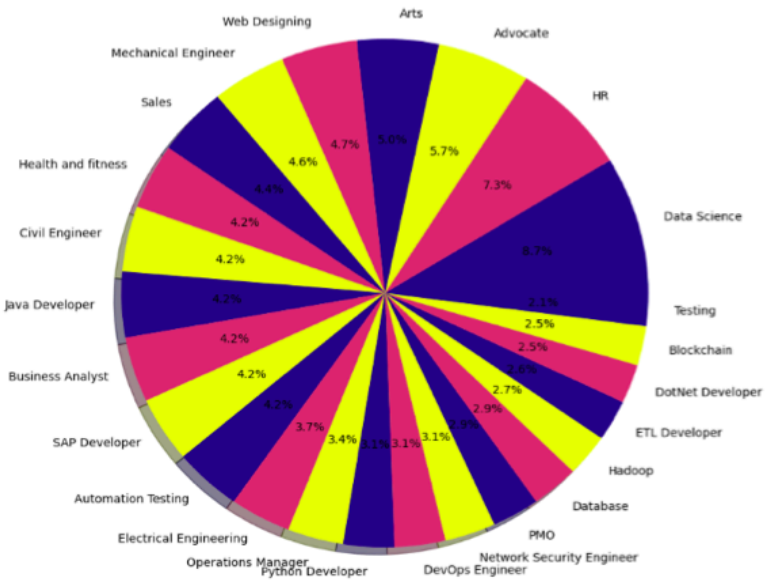


This visualization is essential for identifying any category imbalances in the dataset, which may need to be addressed through techniques like data augmentation or re-sampling to prevent the model from developing a bias toward over-represented job domains.

Resumes often contain noise in the form of URLs, hashtags, mentions, and special characters, which do not contribute to meaningful categorization. Therefore, a critical data preprocessing phase was implemented to clean the text data. This involved two key steps: first, the removal of URLs, hashtags, mentions, and non-ASCII characters using regular expressions; and second, the replacement of punctuation with spaces to prevent tokenization errors during later stages. Following this cleaning, the categorical variable, 'Category,' was transformed into a numerical form using label encoding, where each unique job title was assigned a unique integer ID. This process prepares the data for the vectorization stage. The results of these initial data handling and exploration steps provide a clear picture of the dataset's structure, as further illustrated in Fig. 2, which presents the distribution analysis of the resume categories.

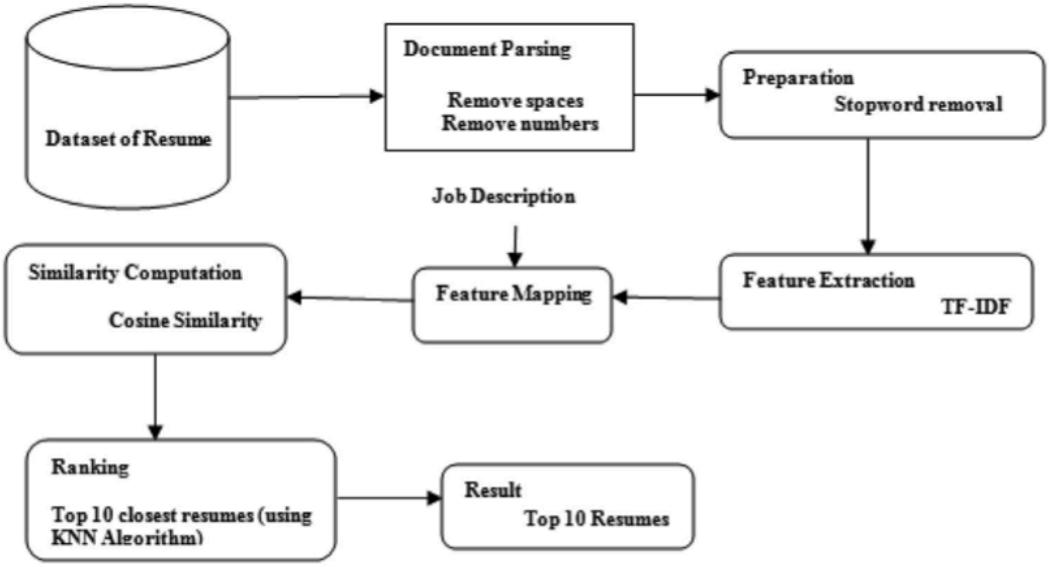
Figure 2 shows the category distribution after preprocessing and label encoding.

9



With the data properly prepared, the proposed system's framework can be implemented. While some systems begin with an initial candidate screening phase using a multiple-choice question (MCQ) test to assess foundational knowledge and prevent malpractice via face detection (Tejaswini et al., 2022), our core framework focuses on the resume itself. Fig. 3, which details the framework of the resume screening and ranking system,

Figure 3 illustrates the full workflow of the proposed resume screening and ranking model.



Illustrates the end-to-end process. The process commences by parsing the resumes from the dataset to remove noise such as white spaces, numbers, and stop words like "and" or "or." The text in both the resumes and the job description (JD) provided by the recruiter is then converted into numerical vectors using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. This is a widely used method for determining word frequencies, where TF-IDF scores aim to emphasize words that are common in a specific text but rare across all texts, making them statistically significant indicators of relevance.

After vectorization, the system needs to assess the similarity of every resume with the job description in order to rank candidates. Saatçı et al. (2024, pp. 121–140) advocate for the application of NLP in resume evaluation by demonstrating how resume details can be analyzed to identify pertinent skills and contrast them with job criteria. This reinforces the decision to convert resumes and job descriptions into numerical formats prior to computing similarity scores

This is a pivotal methodological choice, and the selected metric has a profound impact on the results. A common measure is cosine similarity, which calculates the cosine of the angle between two vectors.

Fig. 4, which shows the general form of cosine distance/similarity in equation:

$$\cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

This helps visualize this concept. This metric is effective because it measures orientation rather than magnitude, making it suitable for comparing documents of different lengths. A score of 1 indicates perfect alignment, while 0 signifies no similarity. However, some research suggests that for technical roles where the presence of a skill is more important than its frequency, Jaccard similarity may be more accurate (Saatçı et al., 2024). Jaccard similarity focuses on the presence or absence of skills (the intersection over the union of skill sets), which prevents a longer resume with repeated keywords from being unfairly favored over a more concise but relevant one. In our proposed system, after calculating the similarity scores, the K-Nearest Neighbors (KNN) algorithm is applied to identify the resumes that are the closest match to the JD provided by the recruiters, effectively generating a ranked list of the most suitable candidates.Nonetheless, newer systems are progressively utilizing semantic similarity and representations based on deep learning. Tian et al. (2023, pp. 202–222) employed LSA, BERT, and SVM in the creation of a recruitment system, whereas Li et al. (2025, pp. 1–24) suggested a model utilizing generative pretrained transformers alongside hierarchical graph neural networks for aligning resumes with job opportunities. These research findings indicate that upcoming resume filtering systems could enhance by integrating conventional keyword-focused techniques with semantic and graph-oriented approaches.

Figure 5. The intersection (A∩B)(A∩B) contains common elements between both sets, while the union (A∪B)(A∪B)represents all distinct elements across the two sets:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

The KNN classifier was trained using the TF-IDF vectorized data, with the dataset split into training and testing sets using an 80-20 ratio. This ensures the model is trained on a large portion of the data while still reserving a subset for validation. The model's performance was then evaluated on the test set by calculating the accuracy score, which is the ratio of correct

predictions to the total number of predictions made. The proposed system achieved an impressive accuracy of 98%, demonstrating its effectiveness in categorizing resumes into their respective job categories (Pahilwan et al., 2025).

While technically effective, such systems are not without significant ethical risks. The most widely discussed challenge is algorithmic bias, which is often not a flaw in the AI itself but a reflection of historical human prejudices embedded in the data it learns from. This bias is particularly acute for marginalized groups, including people with disabilities, who are already twice as likely to be unemployed and thus underrepresented in historical data of successful employees (Nugent & Scott-Parker, 2022). An AI system trained on this biased data might unfairly penalize candidates whose profiles contain "unusual" features, such as a career gap or a non-traditional educational path, even if these features are irrelevant to job performance. Furthermore, using an AI tool as a sole gatekeeper without considering the likelihood that an employer would make a post-offer adjustment is inherently discriminatory. For example, a qualified, visually impaired cybersecurity expert is only the best candidate if the employer enables them to use specialized software. Expecting disabled job seekers to navigate standardized processes without accommodation is analogous to asking a wheelchair user to take the stairs to the interview room (Nugent & Scott-Parker, 2022).

From a theoretical standpoint, the proposed system is a classic example of a Content-Based Recommendation (CBR) system (Freire & de Castro, 2021). It matches items (candidates) to users (jobs) by comparing their content features (skills, experience, as extracted via NLP and TF-IDF). However, the limitations of CBR, such as its reliance on the quality of features and the "cold-start" problem (where it's hard to recommend new items), suggest that a hybrid approach could be more robust. Future enhancements could integrate Collaborative Filtering (CF), which uses user behavior (e.g., which candidates similar recruiters have hired) to make recommendations, or Knowledge-Based Recommendation (KBR), which uses deep domain expertise and ontologies to create more nuanced matches (Freire & de Castro, 2021). The method of matching a candidate's CV with a job description is analogous to a recommender system, a concept first developed by Resnick et al. and now widely implemented across various sectors (Tejaswini et al., 2022).

Finally, implementing such a system faces practical hurdles beyond the technical design. The financial cost is a primary barrier, with a survey indicating that 68% of HR professionals have not implemented an AI recruitment system due to insufficient budget (Chen, 2023). There are also critical data privacy concerns when collecting and processing vast amounts of personal information from numerous sources, which must comply with a complex web of regulations. The role of the human recruiter is also evolving; rather than being replaced, their role shifts to strategic oversight, validating AI recommendations, and focusing on high-value tasks like

assessing cultural fit and building relationships areas where AI currently falls short. The most effective systems are those that foster a cooperative human-AI relationship, where AI handles data-driven tasks and humans provide nuanced judgment and ethical oversight (Chen, 2023).

3.Research Gap

Recent studies show that although AI is becoming a central part of modern recruitment, it still brings major challenges especially when it comes to fairness and bias. Chen et al. (2023) explain that AI hiring systems often inherit the same problems found in traditional recruitment. Because these models learn from past hiring data, they can unintentionally reproduce gender, racial, or age-related biases. Their review covers both technical solutions, like using fairness-aware algorithms, and organizational strategies, such as increasing human oversight. Still, they point out that real progress is limited by the lack of strong governance rules, transparency guidelines, and accountability measures.
Similarly, Albaroudi et al. (2024) dive deeper into the technical methods used to reduce bias, including pre-processing, in-processing, and post-processing approaches. While these techniques look effective in controlled research environments, the authors argue that things become far more complicated in real hiring settings. The everyday messiness of recruitment—like unstructured CVs, vague job descriptions, and long-standing organizational habits makes it difficult for AI systems to operate fairly. This issue is worsened by the fact that there are no consistent evaluation frameworks or standard datasets to measure fairness across different tools.

Although recent studies propose NLP- grounded capsule webbing, semantic matching, and bias-mitigation styles, several gaps remain. First, numerous systems are estimated substantially through delicacy or ranking criteria without sufficient fairness auditing(Fabris et al., 2025, pp. 1 – 54; Wilson et al., 2021, pp. 666 – 677). Second, studies on disability bias show that indeed advanced GPT- grounded webbing systems may disadvantage aspirants when disability- related information appears in resumes(Glazko et al., 2024, pp. 687 – 700). Third, aspirant- perception studies show that campaigners may mistrust completely algorithmic hiring procedures when they believe the system can not fete their oneness(Lavanchy et al., 2023, pp. 125 – 150). thus, unborn exploration should estimate capsule filtering systems not only by specialized performance, but also by fairness, transaprency, beginner commerce, and seeker trust.

In the end, both studies emphasize that addressing AI bias in recruitment requires more than just technical fixes. Chen et al. (2023) highlight the need for collaboration between HR teams, data scientists, and policymakers, so that ethical considerations are part of the entire design and deployment process. Albaroudi et al. (2024) echo this and stress that future research must focus

on building transparent and practical AI systems that can achieve fairness not only in theory but also in actual hiring environments.

4. Methodology

This study is an experimental research method to evaluate the automated applicant's resume screening by using Natural Language Processing and machine learning techniques. The methodology is consistent with the research aim to explore how AI-powered applicant screening can contribute to resume filtering considering efficiency, transparency, and fairness. It has been shown in previous research that manual screening of resumes is time-consuming, inconsistent, and has cognitive bias, while AI-based screening can perform resume parsing, extraction, and evaluation tasks in a more efficient way (Nyathani, 2022; Pahilwan et al., 2025).

The practical implementation was conducted using the resume dataset Resume.csv. The dataset contains 2484 resumes from 24 job categories and has the following columns: ID, Resume_str, Resume_html, and Category. The dependent variable was the resume category. The main input feature was the resume text. The first exploration showed that the dataset was imbalanced. For example, the two largest categories, Information Technology and Business Development, had 120 resumes each, while the smallest BPO category had only 22 resumes. Due to this imbalance, the evaluation was not only based on accuracy, but also the macro-F1 score was used to evaluate model performance over all categories.

The initial stage of data preparation was to clean the resume text. Resumes are often filled with irrelevant information like URLs, emails, special characters, numbers, formatting symbols, and inconsistent spaces. Therefore, preprocessing was applied to remove noises and to standardize the text before modeling was performed. This is in line with the research paper’s explanation that NLP-based resume filtering systems process unstructured resume text by converting it into structured data by identifying relevant elements like skills, qualifications, experience, and job-related information (Pahilwan et al., 2025).

We preprocessed the dataset and split it into training and testing sets using an 80/20 stratified split, resulting in 1,987 training resumes and 497 testing resumes. The dataset was not consistently balanced, and stratification was used to preserve the distribution of resume categories in both subsets.

The study utilized Term Frequency–Inverse Document Frequency (TF-IDF) as the feature extraction method. TF-IDF is a method for converting resume text into numeric vectors. In this method, words that are common in a specific resume but less common in the entire set of resumes get more weight. The method also allows for resume categorization and job-description matching, because it highlights the terms that are statistically significant for a given category.

The paper also stresses the significance of TF-IDF in automatic resume filtering, particularly for the purpose of extracting important textual features and matching resumes to job descriptions.

Three modeling approaches were conducted. The first algorithm was K-Nearest Neighbors (KNN) with TF-IDF. The cosine distance was used to classify the resumes based on similarity. The second was Multinomial Naive Bayes with TF-IDF, which was included for comparison purposes, as Naive Bayes is a popular method in text classification. Semantic embeddings were tested using SBERT with KNN as the third classification model. This model was added because semantic embeddings can capture meaning beyond exact word matching, which is important in resumes where candidates may express similar skills using different terms. Unlike TF-IDF, which depends mainly on word frequency, SBERT represents the resume text based on contextual meaning. Therefore, SBERT + KNN was expected to perform better in categories where wording varies but the underlying job experience is similar.

In addition to this classification, the notebook also implemented the function of ranking the resumes. A job description was cleaned with the same preprocessing function, converted into a TF-IDF vector, and then compared with all the vectors of resumes using the cosine similarity. This is in line with the resume screening structure discussed in the research paper, where the resumes and job description are vectorized, compared using similarity computation, and ranked to find the closest candidate matches.

4.1 Business Understanding

The problem of business in this study is the challenge associated with the efficient and consistent screening of resumes. When using conventional recruitment techniques, it involves a human recruiter who goes through resumes manually, comparing candidate qualifications to job criteria and then short-listing suitable candidates. The process may be time-consuming and subject to personal bias. Hence, the purpose of the system presented is to facilitate the recruiters in categorizing resumes by job domain and ranking them based on their relevance to a specific job description.

The idea of this method is to help recruiters when they first look at people who want to work for the company. It does not mean that recruiters will not be needed at this stage. This is what Chen said in 2023. Chen thinks that people should make decisions and check if someone will fit in with the company culture. People should also make sure that everything is fair. Artificial intelligence can help recruiters with tasks that're boring and need a lot of data.

Horodyski found out that recruiters think artificial intelligence tools are helpful because they save time and reduce the amount of work they have to do. However, recruiters are worried that artificial intelligence does not really understand people. This is a problem when it comes to

making decisions about who to hire without any input. Recruiters like the intelligence recruiting tools because they are helpful but they do not want to rely only on artificial intelligence when making hiring decisions. This is in agreement with Chen (2023, pp. 146–147), who contends that AI ought to work alongside recruiters by managing monotonous screening activities, while human recruiters maintain accountability for ultimate decisions and ethical considerations.

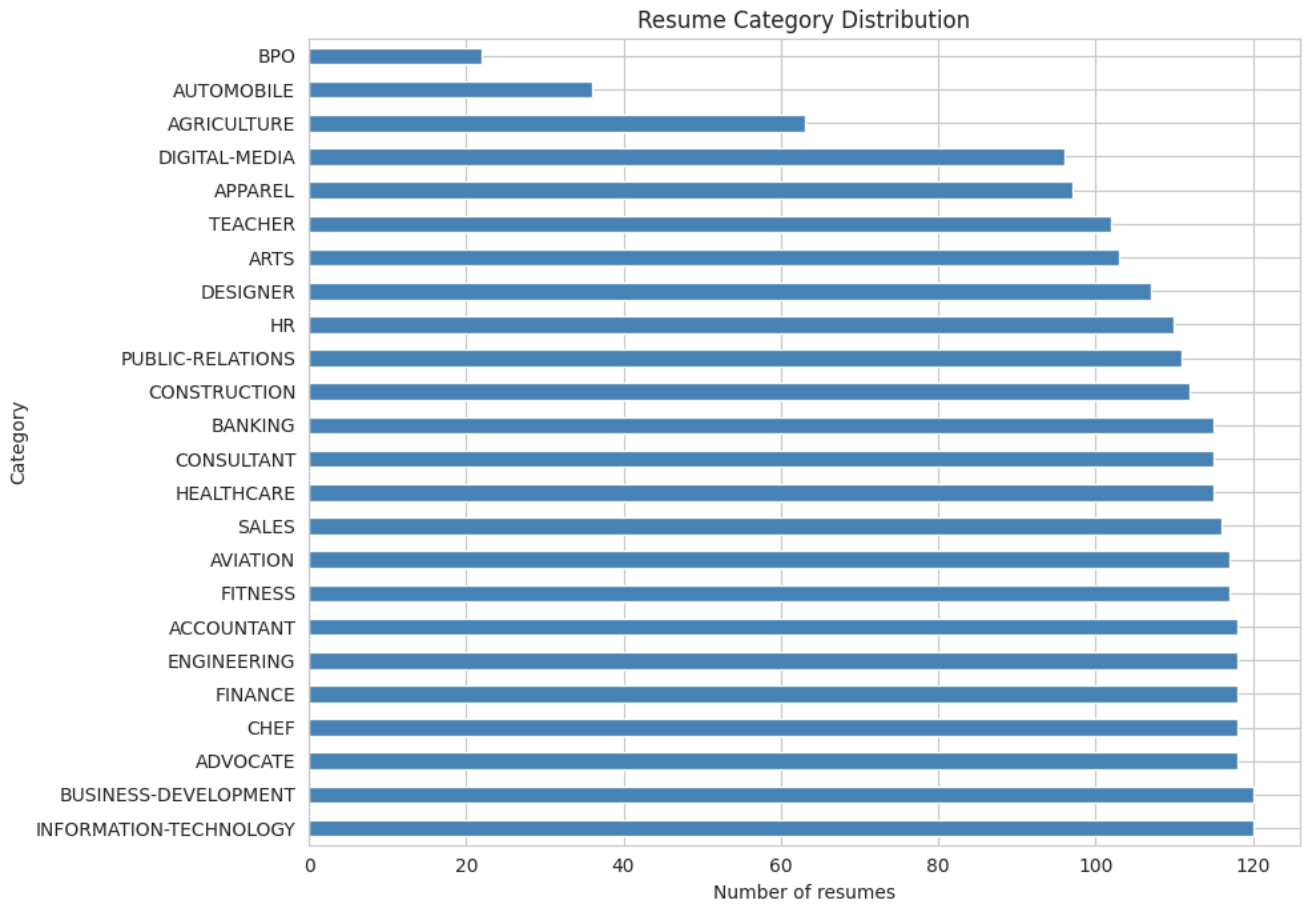
4.2 Data Understanding

The dataset Resume.csv containing 2,484 resumes in 24 different job roles was used. The labels of the category as well as the resumes were the main two columns of the data. The category was the target label for classification, while the resumes acted as input data features.

Preliminary data analysis was performed to identify the imbalance in the dataset. It is important to understand the data because it may impact the performance of the algorithms used. According to Fabris et al. (2025, pp. 1-54), algorithmic hiring systems should be evaluated based on the quality of the dataset, presence of bias, fairness measurements, and the sociological context in which the decisions will be made. Some categories had many resumes while others had fewer resumes.

Understanding this imbalance was important because an imbalanced dataset can cause the model to favor categories with more examples and perform poorly on underrepresented categories. This issue is especially important in recruitment, because biased or incomplete training data can lead AI hiring systems to reproduce unfair patterns found in historical recruitment decisions (Albaroudi et al., 2024; Fabris et al., 2025).

Figure 1. Resume category distribution across the dataset.
This figureshows that the dataset is imbalanced, with some categories containing more resumes than others. This supports the use of macro-F1 in addition to accuracy during evaluation.



4.3 Data Preparation

Before building a model, the resume text went through cleaning and normalization. Most resumes are characterized by noisy elements including links, emails, punctuation, digits, special symbols, non-uniform spacing, and formatting tags. These factors generally do not add value to resume classification or ranking and, therefore, were filtered out during the pre-processing phase. Further, the text was normalized to lowercase to address potential differences in capitalization across terms.

This phase is crucial because resumes are unstructured texts and machine learning algorithms need structured data inputs. Text pre-processing using NLP plays an important role in cleaning up raw text from resumes to prepare for feature extraction and modeling. According to Pahilwan et al. (2025), the application of NLP techniques is essential in automatic resume screening since they enable the conversion of unstructured resume texts into structured data through feature identification, including skills, qualifications, experience, and technical expertise. Similarly, Saatçı et al. (2024) highlight the importance of text pre-processing in resume screening systems before similarity analysis.

For cleaning, an 80/20 stratified split was used to separate the dataset into training and testing sets. This resulted in 497 testing resumes and 1,987 training resumes. The distribution of categories in the training and testing sets was maintained using stratification.

Term Frequency–Inverse Document Frequency (TF-IDF) was then used to convert the cleaned resume text into numerical features. Words that are significant in a particular resume but uncommon in the entire dataset are given more weight by TF-IDF. Because of this, it can be used to find significant terms connected to a job. Although it enables resumes and job descriptions to be represented as equivalent numerical vectors, TF-IDF has been extensively utilized in resume screening and job-matching systems (Pahilwan et al., 2025, pp. 2–5; Saatçı et al., 2024, pp. 121–140).

4.4 Modeling Classification from Resume for Category

Classification of resumes was the first modeling problem. Predicting each resume's job category from its textual content was the task's objective. Put differently, the classification model provides an answer to the query, "Which job category does this resume most likely belong to?"

TF-IDF characteristics were used in the implementation of two classification models. K-Nearest Neighbors (KNN), the first model, classifies a resume by comparing it to the training data's most comparable resumes. Because resume categories may be predicted based on linguistic similarity between resumes, KNN is a good fit for this purpose. The second model was Multinomial Naive Bayes, which is frequently used in text categorization due to its effectiveness with characteristics based on word frequency.

Semantic embeddings were tested using a third classification model, SBERT with KNN. This model was included because semantic embeddings can capture meaning beyond exact word matching, which is important when candidates use different terms to describe similar skills, qualifications, or work experience. Unlike TF-IDF, which mainly relies on word frequency, SBERT represents resume text based on contextual meaning. As shown in the updated classifier comparison figure, SBERT + KNN achieved the strongest performance among the tested models, with the highest accuracy and macro-F1 score. Therefore, SBERT + KNN was considered the best-performing classification model in this experiment and was further analyzed using the confusion matrix to examine category-level prediction performance.

Previous research demonstrating that AI and NLP can help with resume classification, candidate profiling, and application appropriateness prediction supports the use of machine-learning classification in this study (Pahilwan et al., 2025; Will et al., 2023).

This classification task is backed by earlier recruitment studies that utilize machine-learning techniques to classify or assess candidates based on resume content. Tian et al. (2023, pp. 202–222) implemented LSA, BERT, and SVM to create a machine-learning recruiting system.

4.5 Modeling Ranking: Resume against Job Description

Ranking resumes was the next modeling challenge. Unlike classification, ranking does not assign a resume to a certain group. Instead, it ranks the resumes according to their relevancy after comparing each one to a specific job description. "Which CVs are the most pertinent to this job description?" is the question this task attempts to answer.

The job description was cleaned using the same preprocessing algorithm used on the resumes for ranking purposes. The job description was then transformed into a TF-IDF vector. The job description vector was compared to each resume vector using cosine similarity. Resumes with better similarity scores were ranked higher.

This ranking process follows the ideas of resume-job alignment systems, which use similarity assessments to analyze resumes and job advertisements. While Saatçı et al. (2024, pp. 121–140) emphasize matching based on competences between resumes and job specifications, Alsaif et al. (2022, Article 147, pp. 1–3) support this approach by using an NLP-driven bidirectional recommendation system.

This is the same as resume recommendation and matching systems. Resumes and job descriptions are converted into numerical vectors and compared using similarity functions. Alsaif et al. (2022) study bidirectional recommendation systems based on NLP, which recommend jobs to job seekers and resumes to recruiters. Similarly, Saatçı et al. (2024) tackle resume screening using NLP and similarity measures that include comparing resumes with job requirements. Hence, the ranking model in this study operates under the same general principle of text similarity between candidate profiles and job descriptions.

4.6 Evaluation

The classification models were assessed with respect to accuracy and macro-F1 score. Accuracy is defined as the proportion of correctly classified resumes. However, since the dataset is imbalanced, relying on accuracy only may not give a complete picture of model performance. A model could achieve an acceptable level of accuracy by performing well in larger categories but poorly in smaller ones. For this reason, macro-F1 was adopted because it treats all categories equally.

The ranking model was assessed using Precision@5, which indicates how many of the top five retrieved resumes are within the desired job category; this metric was chosen as recruiters in real hiring will care more about ranked candidates than all resumes stored in the database. Evaluation in AI recruitment should not be limited to predicting performance. Automated hiring tools may affect real-life employment opportunities; hence, issues of fairness, transparency, and bias become important considerations. Albaroudi et al. (2024) show that AI hiring systems can propagate or amplify bias when trained on biased data. Fabris et al. (2025) stress that fairness and bias assessment are necessary for algorithmic hiring systems; thus, while this study assesses technical performance through accuracy, macro-F1, and precision@5, an evaluation of fairness is an important requirement for future work. Since recruitment systems influence actual job opportunities, assessment must encompass factors beyond just predictive performance. Alongside accuracy, macro-F1, and Precision@5, forthcoming assessments ought to incorporate fairness auditing and bias testing, as suggested by Albaroudi et al. (2024, pp. 390–395), Fabris et al. (2025, pp. 1–54), and Wilson et al. (2021, pp. 666–677).

4.7 Deployment Considerations

The development of an automatic resume evaluation system is not an easy process, as a lot needs to be considered. In this regard, the proposed system is created to serve as a support for decisions rather than an automated process that takes decisions by itself. It should be noted that recruiters always need to evaluate the recommendations of the system before making their final decision, as this is required because AI can assist in screenings but may not understand certain special cases. Human oversight, clarity, privacy protection, and continuous bias evaluation are all necessary for deployment. This is important because candidates may perceive hiring that is solely algorithmic as unfair (Lavanchy et al., 2023, pp. 125–150), and AI systems may have a negative effect on candidates with disabilities if they fail to take into account reasonable accommodations or experiences related to disabilities (Nugent & Scott-Parker, 2022, pp. 88–90; Glazko et al., 2024, pp. 687–700).

There are several deployment issues that need to be considered. For one, human intervention should be integrated into the software so as to avoid automatically rejecting viable candidates. Secondly, there should be some levels of transparency involved to assist in making the recruiters understand why a resume ranked highly on the ranking list. Thirdly, constant monitoring should be undertaken to check any form of biases or reduction in effectiveness. Fourthly, the issue of data privacy should be considered because the resume contains highly confidential information. Finally, organizations should consider cost, legal compliance, recruiter trust, and organizational readiness before adopting AI recruitment tools.

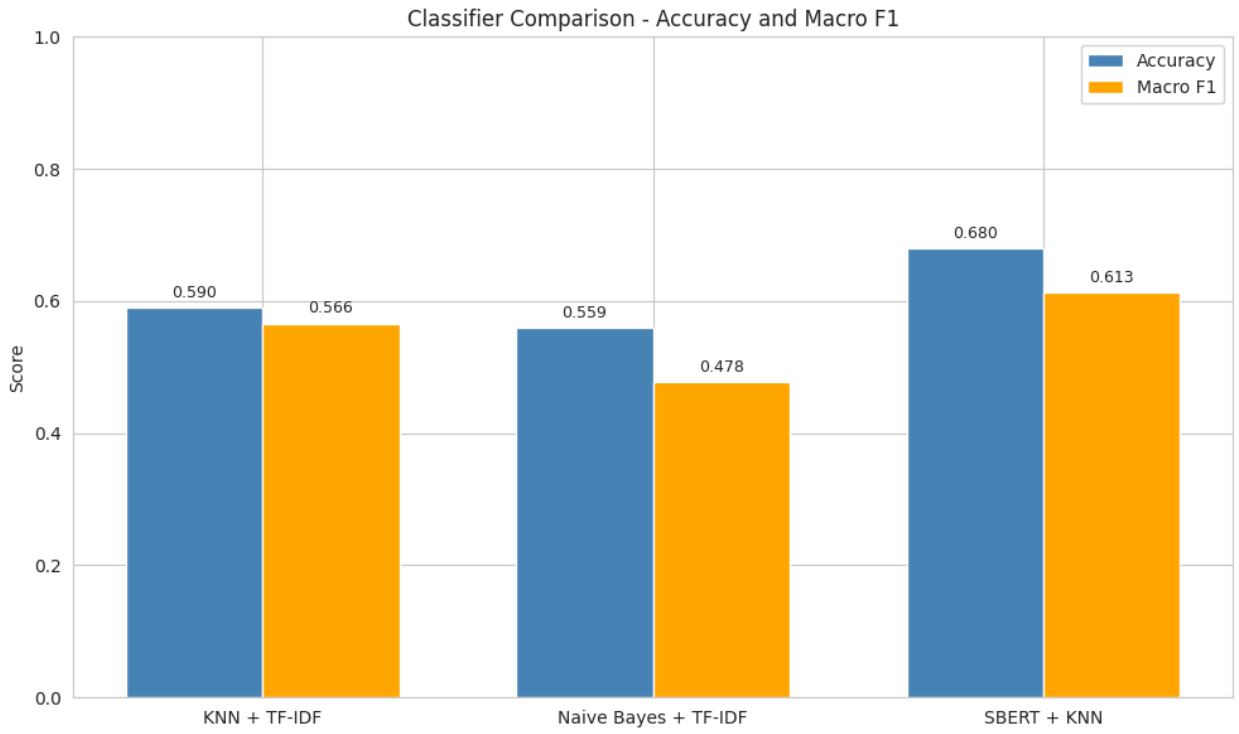
These concerns have been validated in contemporary research on AI recruitment. For example, Chen (2023) suggests that recruiters should work with AI systems instead of being replaced by them. Similarly, Nugent and Scott-Parker (2022) express the possibility that the use of AI in recruiting could lead to a biased selection against people with disabilities, particularly if standardized testing procedures ignore reasonable accommodation. These are also among the concerns identified by Roemmich et al. (2023).

5. Results

On the 24-category resume dataset, the experimental findings show that the built models performed moderately in terms of classification. The accuracy and macro-F1 score of the KNN + TF-IDF model were 0.5614 and 0.5349, respectively. The Naive Bayes + TF-IDF model has a lower macro-F1 score of 0.4857 but slightly higher accuracy of 0.5654.

Figure 2. Classification model comparison using accuracy and macro-F1.

The figure compares the performance of three classification models: KNN + TF-IDF, Naive Bayes + TF-IDF, and SBERT + KNN. The results show that SBERT + KNN achieved the best overall performance, with the highest accuracy score of 0.680 and the highest macro-F1 score of 0.613. KNN + TF-IDF achieved moderate performance with an accuracy of 0.590 and a macro-F1 score of 0.566, while Naive Bayes + TF-IDF achieved lower balanced performance with an accuracy of 0.559 and a macro-F1 score of 0.478. Since macro-F1 is important for evaluating performance across imbalanced resume categories, the results indicate that SBERT + KNN was the strongest and most balanced classification model in this experiment



Since macro-F1 provides a better evaluation for imbalanced datasets, the results indicate that SBERT + KNN is the most balanced and effective model, achieving both the highest accuracy and the highest macro-F1 score compared with KNN + TF-IDF and Naive Bayes + TF-IDF.

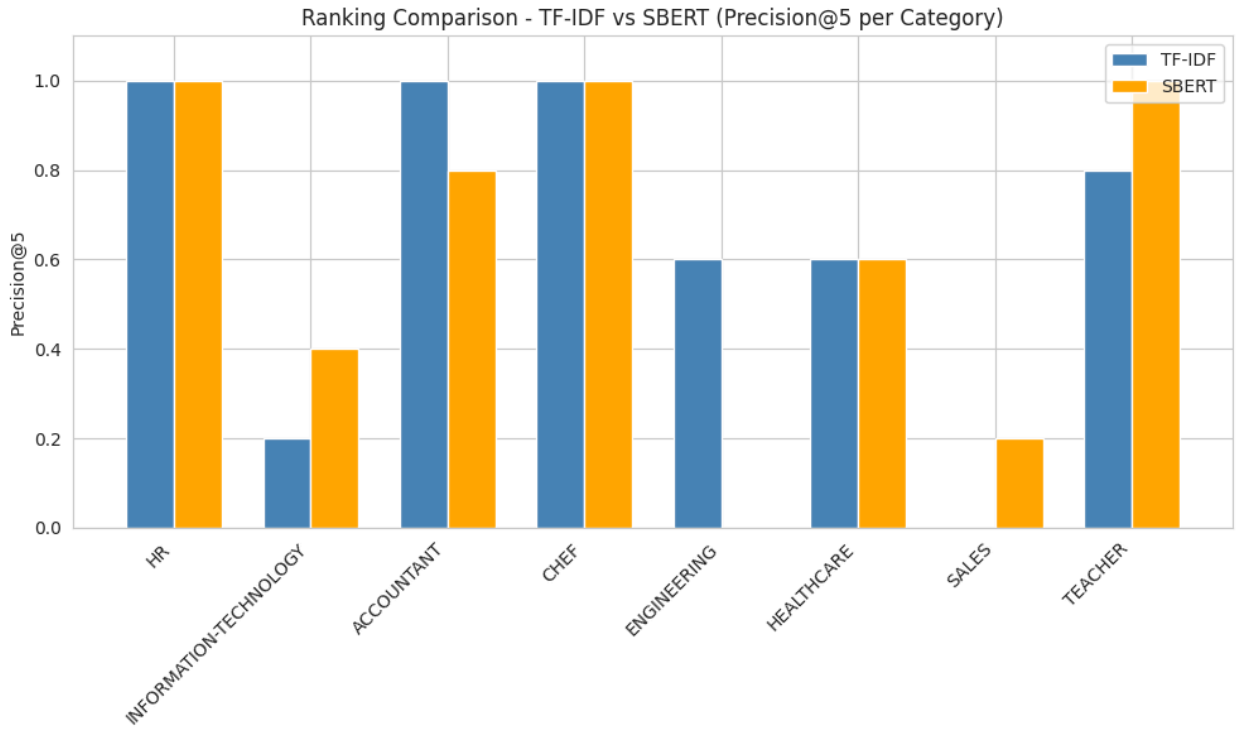
Compared with the classification experiment, the ranking experiment evaluated how well the system could retrieve resumes that matched specific job descriptions. The ranking results show that TF-IDF and SBERT performed differently across job categories. TF-IDF was effective when the job description contained clear and direct keywords, while SBERT was more useful in some cases where resumes and job descriptions expressed similar meaning using different wording. This supports the idea that similarity-based screening can help recruiters identify relevant resumes for shortlisting. However, the variation across categories also shows that ranking performance depends on the wording of the job description and the nature of the job category. Therefore, the ranking system should be used as a recruiter-support tool rather than as a final automated hiring decision.

Eight sample job descriptions from different categories were used to evaluate the ranking models using Precision@5. Precision@5 measures how many of the top five retrieved resumes belong to the expected job category. The results show that ranking performance varied across categories and between the two methods. TF-IDF achieved strong results in categories with clear and specific keywords, such as Human Resources, Accountant, Chef, Engineering, Healthcare, and Teacher. However, its performance was weaker in categories such as Sales and Information Technology, where the wording may be broader or less direct. SBERT improved performance in

some of these categories, showing that semantic similarity can help when resumes and job descriptions use different terms to express similar skills or experience. Overall, the ranking results suggest that both TF-IDF and SBERT have value, and a hybrid approach may provide stronger resume-matching performance.

Figure 4. Resume ranking performance using Precision@5 across job categories.

The figure shows that TF-IDF and SBERT ranking performed differently across job categories. TF-IDF worked better when job descriptions contained clear keywords, while SBERT improved some categories where semantic meaning was more useful than exact word matching. This suggests that both methods have value, and a hybrid approach may improve resume ranking results.



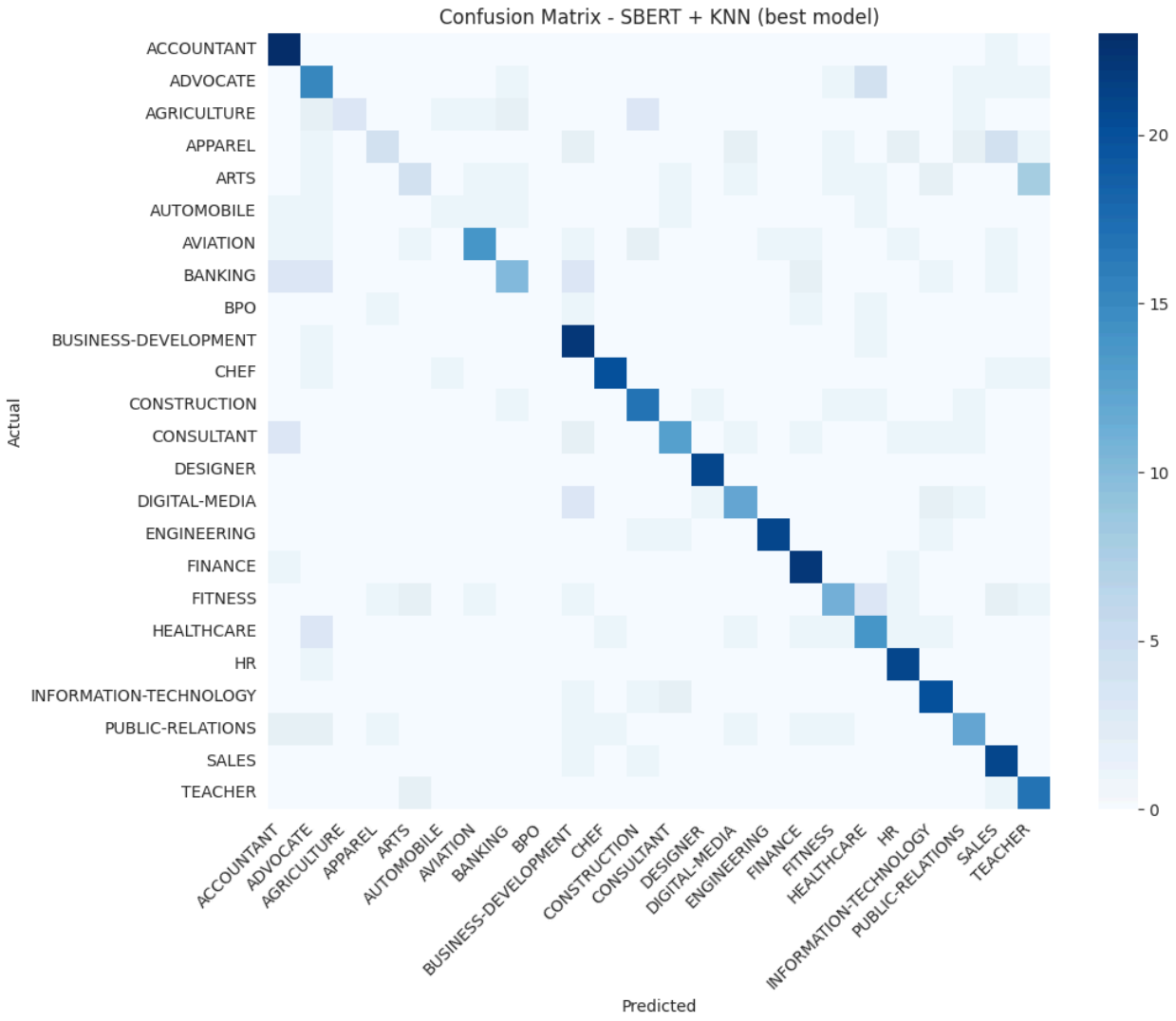
Although SBERT + KNN was originally added as an additional semantic classification model, it was successfully evaluated in the updated experiment and achieved the strongest overall performance. The comparison figure shows that SBERT + KNN reached the highest accuracy and macro-F1 score among the three tested models. This indicates that semantic embeddings improved resume classification by capturing contextual meaning rather than relying only on exact keyword matching. Therefore, SBERT + KNN was considered the best-performing classification model and was further analyzed using the confusion matrix.

Overall, the results indicate that the system has potential for both resume classification and resume ranking, but its performance differs depending on the task. For classification, SBERT + KNN achieved the strongest results, with 68.0% accuracy and 61.3% macro-F1, showing that semantic embeddings improved performance compared with TF-IDF-based models. For ranking, the results varied across job categories, as TF-IDF performed better when job descriptions contained clear keywords, while SBERT was more useful when semantic meaning was needed. This suggests that the system is more suitable as a recruiter-support tool that helps shortlist and organize candidates, rather than as a fully automated final decision-maker. This outcome aligns with the study's discussion that AI recruitment systems can support resume scanning, applicant ranking, and skill matching, but they still face challenges related to accuracy, transparency, fairness, and ethical use.

6. Discussion

The results provide important insights into automated resume screening. The updated classification results show that SBERT + KNN performed better than the TF-IDF-based models, achieving the highest accuracy of 68.0% and the highest macro-F1 score of 61.3%. This suggests that semantic embeddings helped the model understand resume meaning beyond exact keyword matching. However, the confusion matrix also shows that some categories were still misclassified, especially when job domains shared similar skills or vocabulary. Therefore, although SBERT + KNN was the best-performing model, the system should still support recruiters rather than replace human judgment.

Figure 3. Confusion matrix for the best classification model, KNN + TF-IDF.



The confusion matrix shows how the best model classified resumes across the 24 job categories. The diagonal values represent correct predictions, while off-diagonal values show category confusion. The darker spots on the diagonal represent those resumes that were classified correctly by the algorithm to the correct categories; however, the light-colored areas outside the diagonal represent those cases where certain categories were mistaken as other job types due to the similarity of the job fields' vocabulary and/or skill set. These findings are consistent with the quantitative data obtained from the comparative table of the models.

Another important distinction can be observed between KNN + TF-IDF and Naive Bayes + TF-IDF. KNN + TF-IDF performed better than Naive Bayes in both accuracy and macro-F1 score, achieving an accuracy of 0.590 and a macro-F1 score of 0.566, compared with Naive Bayes, which achieved an accuracy of 0.559 and a macro-F1 score of 0.478. This suggests that KNN + TF-IDF produced more consistent performance across the different resume categories. Since the dataset includes categories with varying numbers of resumes, macro-F1 is a more

useful metric than accuracy alone because it gives a better indication of how well the model performs across both large and small categories.

A major finding from the experiment is that the achieved classification performance was lower than the high accuracy values reported in some previous studies. In this study, the best-performing model was SBERT + KNN, which achieved an accuracy of 0.680 and a macro-F1 score of 0.613. This does not mean that previous studies were incorrect, but it shows that model performance can vary depending on dataset size, category distribution, preprocessing choices, feature representation, and evaluation method. This supports the research gap identified in the report, which explains that although AI recruitment systems may show improved technical performance under controlled conditions, it remains unclear whether they perform consistently and fairly in real organizational hiring contexts (Chen, 2023, p. 9)

The ranking outcomes show more potential. In categories like HR, Accountant, Chef, and Teacher, which have different and specific keywords, TF-IDF ranking is achieved exceptionally well. This illustrates how effective TF-IDF is in matching resumes to job descriptions. The poor outcome for sales, however, indicates that TF-IDF has difficulty when job descriptions contain generic terms that are used in a variety of résumé categories. This limit is in contrast to the literature that sheds light on how content-based systems might overlook qualified applicants or rely too heavily on basic levels of text similarity due to insufficient analysis and over-specialization (Freire & de Castro, 2021).

This is especially important because automated screening systems can reproduce or amplify bias when they are trained on biased or incomplete data. Albaroudi et al. (2024, pp. 390–395) explain that AI hiring systems require continuous bias monitoring because they may reinforce discriminatory patterns related to protected characteristics such as gender, race, and age. Similarly, Nugent and Scott-Parker (2022, pp. 88–90) highlight that people with difficulties may be disadvantaged when AI systems rely on historical hiring data or standardized screening procedures that do not consider reasonable accommodations. Therefore, even though SBERT + KNN achieved the strongest technical performance in this study, its results should not be interpreted as proof of fairness. Human oversight, transparency, and fairness testing remain necessary before using automated screening systems in real recruitment contexts.

7. Conclusion

7.1 Summary

This study involved the use of artificial intelligence and machine learning tools in automated applicant screening and resume matching. This study integrated the theoretical aspect of AI in recruitment with an actual implementation involving a dataset consisting of 2,484 resumes grouped into 24 different job categories. The model made use of text processing, term

frequency-inverse document frequency vectorization, classification, and cosine similarity in evaluating resume categorization and matching.

The outcomes indicate that the classification experiment achieved the strongest performance when semantic embeddings were used. SBERT + KNN outperformed the TF-IDF-based models, achieving the highest accuracy of 68.0% and the highest macro-F1 score of 61.3%. In comparison, KNN + TF-IDF achieved 59.0% accuracy and 56.6% macro-F1, while Naive Bayes + TF-IDF achieved 55.9% accuracy and 47.8% macro-F1. These results suggest that SBERT + KNN was the most effective and balanced classification model in this study. However, the ranking experiment showed mixed results across job categories, with TF-IDF and SBERT each performing better in different cases depending on the wording of the job description and the specificity of category-related keywords. Therefore, the system shows potential as a recruiter-support tool, but further testing with real job descriptions and fairness evaluation is still needed before practical deployment.

The results found in this document validate the thesis statement of the paper: AI can make resume screening more efficient if used responsibly. AI systems will allow HR managers to handle a high number of applications, standardize preliminary candidate selection, and screen out individuals who may have relevant qualifications faster. On the other hand, the document highlights that using AI for recruitment is associated with issues of ethics, fairness, bias, data protection, and accountability (Albaroudi et al., 2024; Chen, 2023; Roemmich et al., 2023).

7.2 Limitations

The current research faces several limitations that should be considered when interpreting the findings. First, the dataset used in this study was imbalanced, as some resume categories had a much higher number of samples than others. This imbalance may have affected the learning process of the models, as categories with more resumes could have been easier for the algorithms to recognize, while categories with fewer samples may have been more difficult to classify correctly. As a result, accuracy alone cannot provide a complete or reliable evaluation of model performance, because a model may achieve reasonable accuracy by performing well on the larger categories while still performing poorly on smaller or underrepresented categories. For this reason, macro-F1 was also used as an evaluation metric, since it gives equal importance to all categories and provides a more balanced view of the model's performance across the full dataset. Second, although the study included SBERT + KNN as a semantic classification model and this model achieved the strongest overall performance, the experiment was still limited to a small number of classification approaches. The study did not test more advanced transformer-based models, fine-tuned language models, or different embedding techniques that may produce stronger results. In addition, the KNN model was not deeply optimized using a wide range of K values or distance metrics. Therefore, the results may improve in future work if additional semantic models and larger balanced datasets are used. This limitation shows that while the

current system demonstrates useful potential for automated resume screening, its performance should be viewed as an initial experimental result rather than a final optimized solution.

Third, the job descriptions used for the ranking experiment were manually created for testing purposes, which means they may not fully represent real job descriptions written by recruiters in actual hiring situations. Real job descriptions usually include more detailed requirements, different wording styles, seniority levels, soft skills, technical skills, and company-specific expectations. Therefore, the ranking results should be interpreted carefully, because the visual comparison shows that TF-IDF and SBERT performed differently across job categories. TF-IDF performed better in some categories where the job descriptions contained clear and direct keywords, while SBERT performed better in categories where semantic meaning was more useful than exact keyword matching. This indicates that neither ranking method can be considered universally superior based only on the current experiment. Fourth, although SBERT was successfully evaluated in the updated study and SBERT + KNN achieved the strongest classification performance, the ranking results still varied across categories. This shows that semantic embeddings improved the classification task, as shown in the classifier comparison figure, but further testing is still needed to confirm their effectiveness for resume ranking in real recruitment contexts. Future work should therefore test the system using real recruiter-written job descriptions, larger and more balanced datasets, different semantic embedding models, and hybrid approaches that combine TF-IDF keyword matching with SBERT semantic similarity

Finally, the study did not include fairness or bias testing, which is an important limitation because the current visuals only evaluate technical performance. The classifier comparison figure shows that SBERT + KNN achieved the strongest accuracy and macro-F1 score, and the confusion matrix shows how well the model classified resumes across categories. However, these visuals do not show whether the model treated different demographic groups fairly. This means that strong classification performance should not be interpreted as proof that the system is unbiased or safe for real hiring use. Previous research highlights that AI recruitment systems can reproduce or amplify discrimination when they are trained on biased historical data, especially against protected groups such as applicants affected by gender, race, age, disability, or non-traditional career backgrounds (Albaroudi et al., 2024, pp. 390–395; Nugent & Scott-Parker, 2022, pp. 88–90). Roemmich et al. (2023, pp. 109:5–109:6) also emphasize that many AI hiring tools lack transparency regarding dataset selection, model validation, and legal compliance. Therefore, although the results support the usefulness of automated resume screening as a recruiter-support tool, future versions of the system should include fairness auditing, bias testing, human review, and transparency mechanisms before being applied in real recruitment contexts.

7.3 Future Work

Future work should extend this study in several directions. First, the SBERT + KNN model should be further optimized because the updated classification results show that it achieved the

strongest performance among the tested models. Since SBERT uses semantic embeddings, it was able to capture contextual meaning in resume text rather than depending only on exact keyword overlap. This made it more effective than the TF-IDF-based models, as shown in the classifier comparison figure where SBERT + KNN achieved the highest accuracy and macro-F1 score. Future experiments should therefore test different K values, distance metrics, embedding models, and hyperparameter settings to determine whether the performance can be improved further. In addition, the confusion matrix should be analyzed in more detail to identify which resume categories are still being misclassified and whether additional preprocessing, class balancing, or category-specific training could reduce these error.

Second, further research should use actual job postings written by recruiters and, if possible, include recruiter feedback to evaluate whether the ranked resumes are practically useful in real hiring situations. This is important because the ranking visuals show that TF-IDF and SBERT did not perform equally across all job categories. Some categories achieved stronger Precision@5 results when the job description contained clear and specific keywords, while others benefited more from semantic similarity when the wording between resumes and job descriptions was less direct. Therefore, testing the system with real recruiter-written job descriptions would provide a more accurate assessment of how well the ranking model works outside the experimental setting. Recruiter feedback would also help determine whether the top-ranked resumes are not only technically similar to the job description, but also relevant, qualified, and useful for actual shortlisting decisions. This aligns with the research problem discussed in the literature review, which emphasizes that AI-based resume screening systems should be evaluated in realistic recruitment pipelines rather than only under controlled testing conditions.

Thirdly, fairness testing should be incorporated into future versions of the system. Although the current visuals show useful technical results, such as the classifier comparison, ranking comparison, and confusion matrix, they only evaluate model performance and do not measure whether the system treats applicants fairly. Future models should therefore be tested for possible bias related to gender, disability, age, ethnicity, educational background, employment gaps, and non-traditional career paths, provided that such evaluation can be conducted ethically and legally. This is especially important because a model can achieve high accuracy or macro-F1 while still producing unfair outcomes for certain applicant groups. To address this issue, future work should use balanced datasets, fairness-aware evaluation metrics, bias audits, and human review. These steps would help ensure that the system does not only perform well technically but also supports responsible, transparent, and fair recruitment decisions.

Finally, future research should examine hybrid methods that combine TF-IDF keyword matching, SBERT semantic similarity, and skills extraction. The current visuals show that each approach has different strengths: SBERT + KNN achieved the best classification performance, while the ranking results varied across job categories depending on whether exact keyword

overlap or semantic meaning was more useful. A hybrid model may therefore produce stronger results by using TF-IDF to capture direct job-specific terms, SBERT to understand contextual meaning, and skills extraction to identify the most relevant qualifications in each resume. Fifth, the project should be expanded into an explainable recruiter-support tool. Instead of only showing a ranked list or predicted category, the system should explain why a resume was classified or ranked highly, such as by highlighting matched skills, relevant experience, important keywords, and semantic similarities with the job description. This would allow recruiters to validate the model's output before making decisions and would make the system more transparent, trustworthy, and practical for real recruitment use. This direction also aligns with the paper's recommendation for scalable, interpretable, and ethically compliant models that can operate effectively in dynamic hiring environments (Albaroudi et al., 2024, p. 402).

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Research Problem

Despite the rapid adoption of automated applicant screening systems that use artificial intelligence (AI) and machine learning for resume filtering, current approaches face major limitations in fairness, transparency, and real-world effectiveness. Research demonstrates that although AI-enabled recruitment systems promise to improve hiring efficiency, they often reproduce discriminatory patterns linked to gender, race and other protected attributes, it's a consequence of biased training data and opaque decision-making processes. This research concentrates on algorithmic accuracy or bias detection using small, synthetic datasets rather than live, large-scale recruitment pipelines. This means that while technical performance may improve in controlled conditions, it remains unclear whether such systems perform consistently and equitably under real organizational circumstances (Chen et.al, 2023: p.9).

Research indicates that the lack of standardized evaluation frameworks and benchmark datasets makes it difficult to compare studies, replicate findings, or confirm the fairness of AI screening tools. Although several mitigation methods such as pre-processing, in-processing, and post-processing techniques have been developed to mitigate bias, they are rarely tested in real hiring environments with unstructured resumes, diverse job roles, and multiple languages. This raises ongoing questions about whether these systems truly improve efficiency and fairness or unintentionally exclude qualified candidates and reproduce human-like bias. Additionally, limited research has examined how recruiters interact with AI-generated shortlists, how transparent the selection criteria are, and what ethical or legal issues arise from using personal data in these algorithms (Albaroudi, Mansouri & Alameer, 2024 :p.390-391).

Therefore, there is a pressing need to investigate how AI-driven resume filtering systems can be designed and implemented to balance efficiency, fairness, and transparency in real-world recruitment processes. Future work must go beyond algorithmic optimization in controlled settings and include longitudinal field studies, multi-lingual datasets, recruiter-AI interaction assessments and robust ethical governance frameworks for human-machine oversight Chen (2023;p. 7). Albaroudi et al. (2024; p. 402) made a recommendation for scalable, interpretable, ethically-compliant models capable of operating in dynamic hiring environments.

Research Question

How can automated applicant screening systems for resume filtering be optimized to ensure fair, transparent, and efficient candidate selection?